AUTOMATIC SWARA SEGMENTATION AND IDENTIFICATION IN INDIAN CLASSICAL FLUTE MELODIES: A DEEP LEARNING FRAMEWORK WITH COMPARATIVE ANALYSIS OF ACOUSTIC FEATURES

Submitted by

Akash Datta Roni
Reg: 2020331550

Misbah Al Mamun
Reg: 2020331553

Course Code: CSE 800

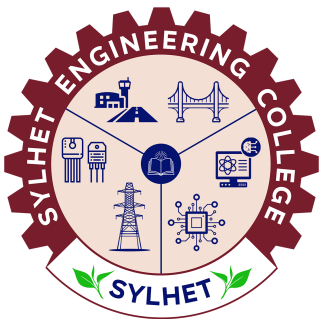
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Supervised by

Md. Abu Naser Mojumder
Associate Professor and Head of the Department
Department of Computer Science and Engineering
Sylhet Engineering College, Sylhet



Sylhet Engineering College
(Shahjalal University of Science and Technology)

Sylhet-3100, Bangladesh

22 August 2026

Candidate's Declaration

This is to certify that the work presented in this thesis entitled, "Automatic Swara Segmentation and Identification in Indian Classical Flute Melodies: A Deep Learning Framework with Comparative Analysis of Acoustic Features", is the outcome of the research carried out by the authors under the supervision of Md. Abu Naser Mojumder, Associate Professor and Head of the Department, Department of Computer Science and Engineering, Sylhet Engineering College, Sylhet.

It is also declared that neither this thesis nor any part thereof has been submitted anywhere else for the award of any degree, diploma, or other qualifications.

Signature of the Candidates

Akash Datta Roni	Misbah Al Mamun
Reg: 2020331550	Reg: 2020331553

Recommendation Letter from Thesis Supervisor

The thesis titled **"Automatic Swara Segmentation and Identification in Indian Classical Flute Melodies: A Deep Learning Framework with Comparative Analysis of Acoustic Features,"** submitted by Akash Datta Roni and Misbah Al Mamun, is being presented under my supervision.

After thoroughly reviewing the thesis, I confirm my approval for its submission for examination.

Md. Abu Naser Mojumder
Associate Professor and Head of the Department
Department of Computer Science and Engineering
Sylhet Engineering College, Sylhet
22 August 2026

Certificate of Acceptance

The thesis is titled **"Automatic Swara Segmentation and Identification in Indian Classical Flute Melodies: A Deep Learning Framework with Comparative Analysis of Acoustic Features"** submitted by **Akash Datta Roni** and **Misbah Al Mamun**; Student ID: **2020331550** and **2020331553**; Session: **2020-2021** has been accepted as satisfactory in partial fulfillment of the requirement for the degree of **Bachelor of Science in Computer Science and Engineering** on August 2026.

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Assistant Professor
Department of Computer Science and Engineering
Sylhet Engineering College

Nayan Kumar Nath — Internal Examiner
Lecturer
Department of Computer Science and Engineering
Sylhet Engineering College

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Abstract

Musical Information Retrieval (MIR) has emerged as an important research area in machine learning, with automatic note segmentation and recognition remaining challenging tasks. While considerable progress has been made for Western instruments such as the piano, comparatively little research has focused on Indian classical music and its instruments, resulting in limited datasets and methodologies for automatic swara transcription. This study proposes an end-to-end pipeline for automatic segmentation and swara classification of monophonic Indian classical flute melodies.

The proposed framework consists of two stages: note segmentation and swara classification. A dataset of 748 individual flute notes was collected through a combination of self-recordings and publicly available recordings. These notes were then used to generate a synthetic melody dataset that closely resembles continuous flute performances. A CRNN with an attention mechanism was employed to identify note boundaries within continuous melodies, while a CNN with an attention mechanism was trained to classify segmented notes using Constant-Q Transform (CQT) features.

To investigate the effectiveness of different acoustic representations, multiple feature extraction techniques, including Log-Mel Spectrogram, CQT, pitch, RMS energy, and their combinations, were evaluated for the segmentation stage. Experimental results demonstrate that CQT-based features achieved the best overall performance. The proposed segmentation model achieved a segmentation accuracy of **92.86%**, while the classification model achieved **97.80%** swara classification accuracy on correctly segmented melodies.

The results demonstrate that accurate end-to-end swara transcription can be achieved using a relatively small dataset when combined with synthetic data generation, careful model optimization, and appropriate feature representation. The proposed framework provides a practical and scalable approach for automatic transcription of monophonic Indian classical flute melodies and has the potential to be adapted to other monophonic musical instruments with minimal annotated data.

Keywords: Musical Information Retrieval; note segmentation; swara classification; Constant-Q Transform; Indian classical flute melodies

NOMENCLATURE

List of abbreviations used in the thesis.

AMT	Automatic Music Transcription
MIR	Musical Information Retrieval
ICM	Indian Classical Music
CRNN	Convolutional Recurrent Neural Network
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
CTC	Connectionist Temporal Classification
CQT	Constant-Q Transform
MFCC	Mel-Frequency Cepstral Coefficients
DSP	Digital Signal Processing
RMS	Root Mean Square (energy)
GPU	Graphics Processing Unit
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative
F1-score	Harmonic mean of Precision and Recall

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Chapter 1

INTRODUCTION

Musical Information Retrieval (MIR) [1] has become an increasingly important research area due to the rapid growth of digital music platforms, machine learning, and artificial intelligence. With the availability of modern MIR systems, computers can automatically analyze, understand and process music content to make them useful in many applications, such as automatic music transcription, music recommendation, source separation, instrument recognition, music retrieval and audio tagging. Understanding musical performances is an important research task, and the need to develop reliable methods to do so is growing due to the rise of intelligent music analysis.

Musical instruments can generally be categorized as either monophonic or polyphonic based on the number of notes they can produce simultaneously. Polyphonic instruments can play more than one note at a time, including the piano, guitar, and harp, whereas monophonic instruments play only one note at a time, such as the flute, trumpet, clarinet, and saxophone. Monophonic instruments do not have any notes overlapping but note boundaries are still a difficult challenge to identify and even note classification from continuous performances is difficult because of variations in note duration, articulation, performer technique, and recording conditions.

Among the various research problems in MIR, Automatic Music Transcription (AMT) [2] has received considerable attention. AMT aims to automatically transcribe an audio recording of a musical performance into notation, without using human intervention [3]. Although significant work has been accomplished for popular Western instruments like the piano, there has been relatively little work on Indian classical instruments. Therefore, there is a lack of public access to classical music datasets, benchmark models and standard evaluation protocols for Indian classical music. This dearth of resources is a major challenge for automated transcription of Indian classical instruments.

The critical challenge in the field of musical analysis is the availability of annotated data, which is one of the main problems in developing machine learning models. Deep learning models generally require large volumes of data that are well-labeled, which can be time consuming and costly to develop. The difficulty is even more pronounced in the realm of Indian classical music, as expert knowledge is frequently necessary to accurately annotate individual swaras and note boundaries. Moreover, there have not been a large number of high-quality recordings of isolated notes or continuous performances compared to other instruments that are more widely studied. This makes it difficult to obtain sufficiently large data sets for supervised learning.

To cope with the above challenges, this research aims to test the feasibility of accurate automatic swara transcription from a small amount of training data. A total of 748 individual flute note recordings were obtained by self-recordings and public sources. A synthetic melody dataset was constructed with 1,000 continuous melodies from the flute for realistic performance

simulation, using these recordings, in order to offer accurate ground-truth for segmentation and classification.

The proposed framework comprises two successive stages, namely note segmentation and swara classification. Firstly, a convolutional recurrent neural network (CRNN) [4] with an attention mechanism is used for note boundary detection in continuous flute melodies. The segmented notes are then classified with a convolutional neural network (CNN) [5] with an attention mechanism [6]. Throughout this research, several acoustic feature representations, including Log-Mel Spectrograms [7], pitch, RMS energy, and the Constant-Q Transform (CQT) [8], were investigated to determine the most effective representation for the proposed system.

Experimental results have shown that CQT-based feature extraction consistently outperforms the other representations used in this study. The proposed segmentation model achieved a segmentation accuracy of 92.86%, while the classification model achieved a swara classification accuracy of 97.80% on correctly segmented melodies. The results show that, with proper feature engineering, synthetic data generation, model optimization and hyperparameter tuning, accurate end-to-end swara transcription can be obtained with a relatively small number of samples.

The main contribution of this research is the development of an end-to-end pipeline for automatic segmentation and swara classification of continuous Indian classical flute melodies. In contrast to the numerous existing methods based on large annotated datasets, the proposed framework shows that competitive performance can be obtained with limited training data through the use of suitable feature representation and attention-based deep learning models along with synthetic data generation. The proposed methodology also offers a practical framework for the extension of automatic music transcription methods to other monophonic musical instruments, which have limited annotated data.

1.1 Motivation

In recent years, Automatic Music Transcription (AMT) has gained considerable interest in the area of music education, digital archiving, music retrieval and intelligent music analysis, among others. In contrast to this, most of the current research has mainly explored Western musical instruments, especially the piano, due to the availability of large-scale annotated data sets and well-developed benchmark models. Conversely, the music of Indian classical instruments, notably the flute, is a comparatively neglected aspect despite its cultural relevance and popular use.

The lack of good quality labeled datasets is one of the key challenges in the research in this field. It takes a lot of time, knowledge and handwork to create annotated musical data, where the boundaries of the notes and swaras need to be identified correctly. This makes it challenging to build deep learning models for Indian classical music, as traditional methods typically require thousands of annotated music recordings to train the model effectively.

In this research, we are motivated to build an efficient and practical automatic swara transcription system which can attain high performance without the need for huge annotated datasets. This work aims to explore whether it is possible to achieve a reliable alternative to real recordings, using synthetic melody generation techniques, suitable feature extraction

methods and attention-based deep learning models. Moreover, the objective of this study is to add to the comparatively small amount of research work about the analysis of Indian classical flute and serve as a platform for further research on other monophonic musical instruments.

1.2 Objectives

The primary objective of this research is to develop an end-to-end deep learning framework capable of automatically segmenting and classifying swaras from continuous Indian classical flute melodies. To accomplish this objective, the following goals were established:

- To collect and prepare a dataset of isolated Indian classical flute notes.
- To generate a synthetic melody dataset that accurately represents continuous flute performances while providing precise ground-truth annotations.
- To develop an automatic note segmentation model capable of identifying note boundaries in continuous audio recordings.
- To investigate and compare multiple acoustic feature representations, including Log-Mel Spectrogram, pitch, RMS energy, and Constant-Q Transform (CQT), for note segmentation and swara classification.
- To design and evaluate attention-based deep learning models for both segmentation and classification tasks.
- To analyze the effectiveness of the proposed framework using quantitative performance metrics and transcription results.
- To investigate whether accurate swara transcription can be achieved using a relatively small amount of training data through careful feature engineering, synthetic data generation, and model optimization.

Chapter **2**

LITERATURE REVIEW

2.1 Introduction

The study of Automatic Music Transcription (AMT) has been ongoing for decades, but most of the published research has focused on music from instruments and traditions that already have extensive annotated collections available. This chapter reviews the research that shaped the design decisions made in this thesis: general approaches to AMT, the particular difficulties associated with monophonic instruments and low-resource corpora, acoustic feature representations used for note-level analysis, deep learning architectures used for onset detection and classification, prior work on Indian classical music, and techniques for generating synthetic training data.

Together, these strands inspire the two main decisions taken in this work, that is, creating a synthetic melodic corpus from isolated flute recordings and using a CRNN-based segmentation stage in combination with a CQT-driven CNN classification stage. The chapter is structured in the following way. In Section 2.2 general AMT frameworks are reviewed, Section 2.3 provides a review of monophonic transcription as well as low-resource instrument corpora, Sections 2.4 and 2.5 review feature representations and neural architectures respectively, and Section 2.6 reviews work specific to Indian classical music. Section 2.7 reviews synthetic data generation and Section 2.8 states the research gap that this thesis addresses.

2.2 Automatic Music Transcription: General Approaches

The task of AMT, to transform the audio signal into a symbolic representation of onset and duration, pitch, and instrument identity and in the case of polyphonic music, separation of sources, is known as one of the long-standing open problems in music information retrieval (MIR), due to the diversity of sub-tasks it encompasses: onset and offset detection, pitch estimation, instrument identification, and source separation for polyphonic material [3]. Following this second overview by the same group [2], the field can be divided into frame-level, note-level, stream-level and notation-level transcription: the latter is all that is required here, since the present work is at the level of notes, not of frames, and does not deal with the stream level, merely with the boundaries and labels. A recent systematic survey of Jamshidi et al. [9] documents the shift of the field from non-negative matrix factorization methods to neural methods, and reports the dominance of convolutional and recurrent neural architectures in published AMT systems, which can be trained end to end on spectrogram-like inputs without hand-designed pitch-tracking rules.

Within AMT, piano transcription has been the most studied task, with thousands of hours of aligned audio-MIDI pairs available for some of the most popular piano datasets, like MAPS and MAESTRO. Hawthorne et al. [10] proposed a novel Onsets and Frames architecture that combines a convolutional-recurrent onset detector with a framewise pitch detector and conditionally feeds the pitch detector on top of the onset detection result, which greatly outperformed previous systems on MAPS in terms of note F1-scores. Even though the instrument, the scale of examples and the type of network used vary significantly, the underlying principle is the same as the two-stage segmentation-then-classification design adopted in this thesis—that is, separating the detection of onsets from the classification of sustained pitches, and then combining the two tasks to produce a better detection of note boundaries than either task would produce individually. But the transfer isn't automatic: The onset head in Onsets and Frames takes advantage of the sharp percussive attack of a struck string, while the flute doesn't develop such an attack. This is discussed again in Section 2.5.

2.3 Monophonic Transcription and Low-Resource Instrument Corpora

Until neural methods became the norm, monophonic pitch estimation was performed using signal-processing algorithms which serve as good baselines. YIN [11] estimates the fundamental period in the time domain from the difference function obtained by its autocorrelation; SWIPE [12] matches a sawtooth-inspired kernel in the frequency domain; and pYIN [13] subsequently put a hidden Markov layer on top of the YIN candidates, resulting in smoother pitch tracks. These methods are correct when given clean sustained tones and require no training data, but they estimate pitch rather than note events; to convert a frame-wise pitch curve into discrete note events, heuristic note segmentation is still necessary and will fail when the note is rapidly articulated or ornamented and when there is expressive pitch movement. Among the reasons that make this thesis different from the approaches that consider segmentation as a post-processing task is that gap between pitch tracking and note segmentation.

Most of the literature of AMT focuses on polyphonic material, where multiple pitches must be separated. Monophonic transcription is simpler in principle, but it is far from solved for instruments that have no large-scale annotated corpora. A very useful reminder that classical DSP techniques still serve as a strong baseline when annotated audio is scarce is provided by Scorpiano, a signal-processing pipeline for monophonic piano transcription developed by Sofronievski and Gerazov [14], which combines onset detection, tempo estimation and pitch tracking and reported accuracy competitive with commercial neural systems while requiring no training data at all. Another approach was chosen by Thomé and Ahlbäck [15], who trained a convolutional-recurrent (ConvLSTM) pitch detector without using an explicit musical language model and found an F-measure of 83% on the 2007 multi-F0 development set from the MIREX competition on a woodwind ensemble recording of a bassoon, clarinet, flute, horn and oboe. The system is polyphonic rather than monophonic, so it is cited here not as a monophonic baseline but as an indication that a learned, timbre-agnostic representation can deal with flute-family material without instrument-specific tuning. Bittner et al. [16] make a similar argument

more directly: their lightweight instrument-agnostic system, trained simultaneously for onsets, multipitch and note activations and specifically motivated by low-resource, instrument-agnostic transcription, uses a compact model built on a harmonic CQT front end that generalises across instrument families at a small cost in accuracy relative to specialised systems.

Two studies specifically address monophonic transcription of underrepresented instruments and can be considered as the nearest parallels to the present study. Loresco et al. [17] created a deep learning system for transcribing the diwdiw-as, a Filipino bamboo panpipe, using a specially created transcription dataset and achieved 96.11% pitch accuracy and 94.07% note-duration accuracy. They found that competitive transcription accuracy is achievable for a folk monophonic instrument, without any access to a MAPS-scale corpus, if the pipeline is constructed based on the existing resources, rather than using them wholesale from piano AMT. Jancovic, Köküer and Baptiste [18], working on ornamented Irish traditional flute recordings, used hidden Markov models over MFCC features, with separate models for notes and for single-note ornaments, and reported that the soft onsets and very short durations of ornaments remain a major source of transcription error even when the underlying pitch content is estimated well. This is the most directly relevant caution for the present study: the instrument family is the same, and the synthetic melody generator used here introduces comparable articulation variation through its transition styles and DSP augmentations. Taken together, these papers frame the problem this thesis addresses: transcription systems for culturally significant but data-poor monophonic instruments cannot simply reuse piano-scale training regimes and must be designed around the data constraints from the outset.

2.4 Acoustic Feature Representations

The accuracy of segmentation and classification is directly related to feature choice and there are repetitions of the comparison of MFCC, Mel-spectrogram and Constant-Q Transform (CQT) representations in the transcription literature. The CQT, in the efficient invertible formulation of Schörkhuber and Klapuri [8], has frequency bins laid out on a geometric grid, meaning that the spacing between semitones is the same throughout the range. There are two consequences relevant to note-level work. First, a change of pitch is a fixed translation along the frequency axis, which suits convolutional kernels that share weights across that axis; this is not possible with a linear or a Mel-scaled spectrogram. Secondly, the constant-Q property provides greater frequency resolution in the low register where the bamboo flute's fundamental frequency is found, but at the expense of coarser time resolution in that register. To be sure, MFCCs were engineered for speech: the cepstral truncation employed to make the compact model for phonetic modelling is purposeful and trades off the fine harmonic detail that contains pitch information.

This is supported by a comparison of empirical data but not made universal. Under matched conditions, several input representations were compared for neural transcription systems by Cheuk et al. [19] and they determined that the type of time-frequency representation has a significant effect on transcription accuracy. Hernandez-Olivan et al. [20] evaluated the performance of a Mel-based Onsets and Frames model against a CQT-based salience model, on the Slakh2100 dataset (consisting of a flute subset), with the conclusion that the CQT-based model deteriorated much less on weaker onset instruments than the Mel-based Onsets and Frames model, and

that training with isolated instrument families further boosted performance. Hebbar and Jagtap [21] studied the effects of the choice of preprocessing and several deep architectures on the task of Carnatic raga recognition in the context of Indian classical music, and thus paved the way for considering the feature representation as a controllable experimental variable rather than a design decision. The counter-example is also illuminating: Wolf-Monheim et al. [22] evaluated spectral and rhythm features for general-purpose music classification and concluded that both Mel-scaled spectrograms and MFCCs outperformed CQT and CENS chromagrams. Combined, these findings suggest that the best representation is task-specific: CQT tends to win where the target is note-level pitch content on a musical scale, and the question of which representation to use should therefore be settled experimentally for the flute rather than assumed. This is the methodology that is followed in Chapter 4 of this thesis.

2.5 Neural Architectures for Onset Detection and Note Segmentation

Onset detection is the sub-task most closely related to the segmentation part of this thesis. Schlüter and Böck [23] demonstrated that a CNN trained on short excerpts of log-Mel spectrograms performed better than the previously state-of-the-art spectral-flux detectors, which had been hand-designed for more than a decade, and also needed significantly less tuning of the manual signal-processing components. Their result is often described as early evidence that, for onset detection specifically, learned convolutional features generalise better than engineered ones. Earlier, Eyben et al. [24] had demonstrated that a bidirectional LSTM trained on an auditory spectral feature could identify onsets across a wide range of instrument timbres without retuning, and that the modelling of temporal context, rather than just the local spectral shape, was useful for this task. Çakır et al. [25] combined the two ideas into a convolutional-recurrent architecture for polyphonic sound event detection, which was subsequently extended to include a frame-level attention mechanism that down-weighted irrelevant or silent frames in follow-up work. This is the same model family, CNN feature extraction, RNN temporal modelling and attention-based frame weighting, as is found in the CRNN-with-attention segmentation model that is used in this thesis. In the same spirit, attention has been applied within AMT itself with Wu et al. [26] translating multi-instrument transcription as an instance segmentation task, where note events are the objects to be segmented, and not the frames to be labelled separately.

A second, and less frequently mentioned, strand is more directly relevant to the flute, since the flute onsets are pitched, non-percussive, and have a slow amplitude rise, as opposed to a sharp transient. In an alternative approach, Stasiak et al. [27] combined multiple onset detection functions by means of a multi-layer perceptron instead of thresholding a single function and tested on seventeen instruments and ensembles. Mońko and Stasiak [28] then tested a CNN onset detector specific to bowed string recordings, the canonical soft onset case, where a CNN trained on mixed material proved to be transferable to solo recordings of a sustained-tone instrument. Mounir et al. [29] took a different approach to the same problem from the feature side, by proposing the NINOS^2 spectral sparsity measure, and demonstrated its superior performance compared to the traditional detectors at the same time as they are failing: sustained-string

instruments, repeated notes, and vibrato passages. The three papers are relevant to this work because they discuss the failure modes of onset-based segmentation: repeated notes, vibrato and slurred onsets, and these three failure modes are present in the synthetic flute melodies employed here, both in the slurred and crossfaded transitions, and in the vibrato augmentation. However, none of the reviewed onset-detection papers test on Indian classical instruments or on a collection of synthetically concatenated recordings of single notes—this is the specific scenario that this thesis addresses.

2.6 Note and Raga Recognition in Indian Classical Music

Computational analysis of Indian classical music (ICM) has been discussed thoroughly by Singh and Biswas [30], who have concluded that the field has focussed on a small set of tasks, raga recognition foremost among them, and that note-level analysis is still underdeveloped. A difficulty specific to this repertoire is tonic (śruti) normalisation. Unlike Western instruments with fixed pitch, the absolute pitch of a particular swara can vary from one recording to another because the performer chooses the tonic (Sa) when performing it. Another approach to the identification of tonic was proposed by Bellur et al. [31] (pitch-histogram method) and by Gulati et al. [32] (group-delay method) who later compared the main ideas on the same set of test data; Singh et al. [33] went the other way around by using a shruti-based chromagram to build the representation of the tonic. Many raga-recognition systems choose not to take this approach, and instead use features that are independent of the tonic, such as chroma or pitch-interval. The segmentation stage of this thesis, however, works on absolute CQT contours, since detection of boundaries is based on the change in the spectrum, and not the identity of the scale degrees; the result is that the classification stage has a dependency on the tonic of the source recordings, which is treated as a limitation and direction for future work in Chapter 6 rather than solved.

At the melodic modelling level, early work by Dighe et al. [34] exploited the occurrence and transition statistics of a swara histogram, with classical classifiers, where it was demonstrated that swara occurrence and transition statistics alone have high discriminative power, even in absence of deep models. More recent systems rely on neural sequence systems: DeepSRGM [35] employs an LSTM based recurrent system to classify and rank Carnatic recordings by raga; PhonoNet [36] is a multi-stage CNN pipeline to identify a Hindustani raga from recordings; Shah et al. [37] report high accuracy on a subset of 10 ragas from the CompMusic collection, but their system consists of CNNs over raw spectrograms. Singh and Arora [38] take the same task, but with explainability as the primary concern, and demonstrate that the regions their model allocates are not tied to a single note, but rather to short melodic phrases, which is a useful reminder for any system, including the system proposed here, that treats the boundaries of the swaras as if they were as distinct as the onsets of notes on the piano. Two studies by Nuttall et al. [39, 40] directly address the note-level, using deep clustering and deep embedding respectively to model the effect of the melodic context on the acoustic realization of a given svara and the effect of the gamaka ornamentation in carnatic music. The conclusion of their work, that a svara does not have a fixed spectral shape, but is highly contextualised by its neighbours, is very relevant to the classification part of this thesis, as the transition-style and

DSP augmentation parameters in the synthetic melody generator also cause a comparable contextualisation.

The most similar system to the current work to the best of our knowledge is the one of Singh et al. [41] which uses a convolutional-recurrent network trained on the CTC objective for Indian classical music transcription. They use a lot of CTC, which means that the labels are not aligned with respect to the frames, but CTC in return means that the model makes a prediction of a sequence of symbols rather than an explicit prediction of onset and offset times. The approach taken in this thesis is the inverse direction, that is, predicting frame level boundaries directly, and it is this that makes the note-duration and timing information recoverable. None of their works nor the literature surveyed above discuss, let alone assess, the performance of the bamboo flute, nor is there any literature that treats frame-level segmentation of continuous melodies into individual swaras.

2.7 Addressing Data Scarcity through Synthetic Data Generation

The synthetic and recombined training data is part of a larger field of work on synthesis and recombination of training data and the strategy for generating the synthetic melody used in this thesis is particularly applicable to Indian classical flute performance since there are few annotated recordings. In the field of music transcription, the most similar dataset is the Slakh dataset [42] which uses virtual instruments to generate a large amount of audio from MIDI transcriptions with automatically generated, exact annotations; this dataset is now a well-known benchmark for multi-instrument transcription. What Slakh proposes as the main idea behind this domain of procedurally generated audio is that, if the ground truth itself is created by the generator and not by a human annotator, then the ground truth is exact by construction and is available at any scale. The main drawback of it, as anyone who has tried it will know, is that the sound of a virtual instrument made using samples is not the same as the sound of a live instrument.

Instead of synthesizing sounds from symbolic input, all of the acoustic units of the corpus are real versions of the target instrument with exactly annotated boundaries, with programmatic control of transitions and DSP augmentations. Work elsewhere in the audio community follows the generative rather than the recombinative paradigm: Ghosh et al. [43] add a text-to-audio diffusion model whose samples are optimized for the audio classification target distribution via preference optimisation, and Ibaraki and Chiang [44] create training data for low-resource speech recognition by augmenting new text with text-to-speech synthesis. Both agree that, although synthetically-generated data can replace limited real data, it is not simply an extension of the size of a data set; neither retains the acoustics of a particular physical instrument in the way that recombination does. This should always be explicitly mentioned, as this will affect the interpretations of the results of this thesis: concatenated melodies will produce a timbre of the source notes but not the continuous control over breath, dynamics and pitch, or the gamaka-like quality, that features a live performance and therefore, accuracy in this corpus is an upper limit of what would be expected on an unconstrained recording. This gap is mentioned in chapter 6

and is bounded by the evaluation on held-out real audio.

2.8 Summary and Research Gap

There are three patterns that run through this body of work. First, there are already a number of good examples of piano AMT, and, increasingly, of multi-instrument settings (polyphonic), though monophonic transcription of non-Western or folk instruments is rather sparse and limited to a few isolated studies, most of which follow a similar approach to that used here, building small custom datasets. Second, the ideas of frame-level attention, CQT input and CNN-RNN fusion transfer naturally to the frame-level Indian classical flute swara localization task, but soft onset instruments like bowed strings have been previously explored in the context of onset detection and polyphonic sound event detection, but not, to the best of our knowledge, in the context of frame-level attention for swaras in continuous Indian classical flute melodies. Third, while many studies have focussed on the raga level classification of the Indian classical music literature, and, more recently, on the context dependent realization of swaras and the transcription of sequences with the help of CTC, the first and the second part have not been integrated in one pipeline and tested on synthetically generated continuous melodies where they can be compared systematically according to their respective features.

The framework that was created in the rest of this thesis is located at the nexus of these three gaps. It makes three contributions, in particular:

(i) it introduces an adaptation of attention-based convolutional-recurrent segmentation, which is successfully used in the general AMT and sound event detection literature, for the relatively under-studied monophonic Indian classical flute, where flute onsets and note boundaries are not accompanied by a percussive transient;

(ii) it compares the results of CQT with Mel-spectrogram, MFCC, pitch and RMS energy features for this specific timbre and task instead of taking a conventional approach to the representation; and

(iii) It presents a recombination-based synthetic melody generator which eliminates the manual annotation burden while retaining all acoustic units in the framework of a real flute recording, and which is instrument-independent in its construction.

Chapter **3**

DATASET PREPARATION

3.1 Dataset Preparation

Data Source. The audio recordings used for training the proposed model were collected from two primary sources: publicly available recordings from YouTube and recordings performed by the authors. The recordings were chosen from YouTube to get good quality low noise samples and the self-recorded flute performances were included to add real environmental noise and recording variations. The joint use of these two sources provided a more diverse dataset and allowed the model to learn from both clean and noisier sound scenarios; the result was a more robust model.

All the audio pieces were then manually classified based on their swara (Sa, Re, Ga, Ma, Pa, Dha, Ni). The recordings were then divided into individual folders with each folder containing audio samples of one swara. The folder-based structure made data management easy and helped with data pre-processing and training of the proposed framework. The folder structure of the dataset is shown below in Fig. 1.

```
recording for paper/
    re/
        re_53.wav
        re_42.wav
        ... (105 more)
    ga/
        ga_38.wav
        ga_35.wav
        ... (104 more)
    sa/
        sa_35.wav
        sa_89.wav
        ... (104 more)
    dha/
        dha_32.wav
        dha_53.wav
        ... (107 more)
    ni/
        ni_22.wav
        ni_34.wav
        ... (105 more)
    pa/
        pa_56.wav
        pa_65.wav
        ... (104 more)
    ma/
        ma_100.wav
        ma_10.wav
        ... (105 more)
```

Figure 1: Folder structure of dataset

The swara class consists of more than 100 audio recordings, which is considered as a sufficiently balanced dataset in order to train a model. Moreover the recordings have different note lengths which enables a variety of natural time variations in the flute performances. The following figures (Figure 2) show the class distribution and the duration distribution of the individual swara recordings.

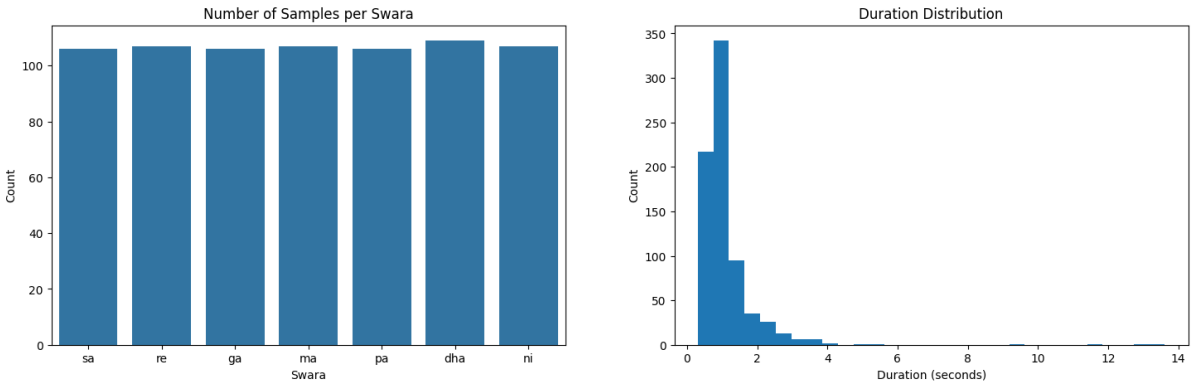


Figure 2: Class Distribution and Duration Distribution of Collected Swaras

Table 1: Individual Swara Dataset

Metric	**Value**
Total Recordings	748
Average Recording per Class	106.86
Number of Swara Class	7
Average Duration	1.137 (sec)
Median Duration	0.968 (sec)
Minimum Duration	0.305 (sec)
Maximum Duration	13.62
Standard Deviation of Duration	1.014 (sec)

Synthetic Melody Dataset Creation. A synthetic melody data set was created by synthesizing a sequence of flute recordings of the seven fundamental swaras (Sa, Re, Ga, Ma, Pa, Dha, Ni) in order to facilitate the evaluation of the proposed swara segmentation and classification framework, in a manner similar to the general recombination strategy employed in other works to derive a large annotated data set from isolated source material [42]. All of the melodies were built as a sequence of individual note recordings, and each note was given a temporal sequence index to maintain its position in the melody. The starting and ending times for each swara were also noted during the generation process to obtain a list of frame-level ground-truth which precisely indicates the temporal boundaries of each note in the synthesized melody. These annotations were stored in a metadata file containing the melody filename, note sequence index, swara label, start time, end time, transition styles and dsp applied. A portion of the melody dataset is presented in Table 2.

Table 2: Synthetic melody dataset

Melody file	**Note_sequence**	**Swara_label**	**Start_time**	**End_time**	**Transition_style**	**Applied_dsp**
flute_melody_0.wav	0	Sa	0	0.4043	phrase_start	speed(1.05)+vibrato+gain(1.02)
flute_melody_0.wav	1	Ma	0.3325	0.9921	slurred_crossfade	pitch(0.17)+speed(1.16)+vibrato+gain(0.63)+noise
flute_melody_0.wav	2	Pa	1.2282	3.4976	articulated_with_breath	pitch(1.06)+speed(0.82)+gain(1.23)+noise
flute_melody_0.wav	3	Ni	3.6154	4.678	articulated_with_breath	pitch(1.38)+speed(1.08)+gain(0.82)+noise
flute_melody_0.wav	4	Ni	4.9613	6.0312	articulated_with_breath	speed(0.99)+gain(1.22)

The synthetic melodies were generated by randomly selecting note recordings from the pool of isolated swara samples. The sequence of swaras and number of notes for each melody and the specific recording for each swara for every melody were programmed optimally, based on the criteria of getting the most diversity and ensuring that all seven swaras were represented throughout the dataset. Because there were several recordings for each swara, the generator selected different source recordings across melodies, preventing repeated use of the same note samples. Additionally, each note in the melody was also given different time values, and melodies had varying lengths and different rhythms. This randomization approach resulted in a diversity of melodic patterns and in accurate ground-truth annotations for all notes. The diverseness of notes per melody and duration is shown in Figure 3.

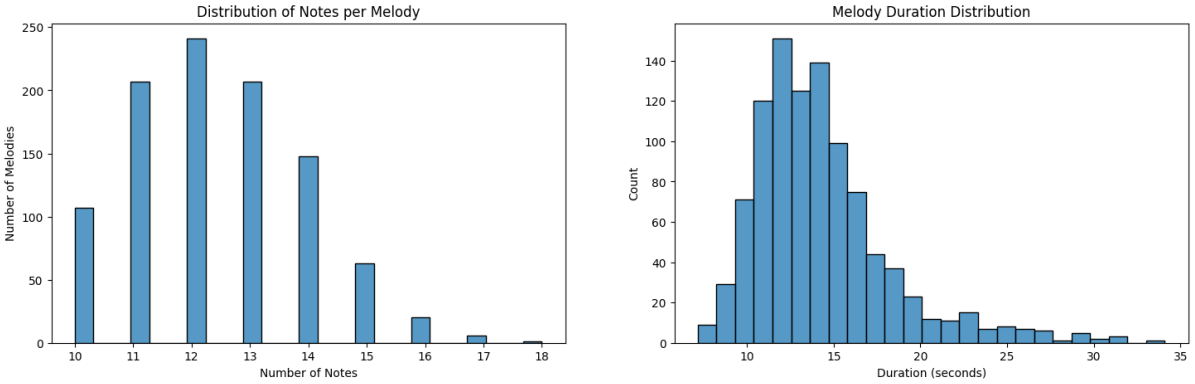


Figure 3: Melody Duration and Swaras used per Synthetic melody

To further improve the realism of the synthesized melodies, multiple transition techniques and digital signal processing (DSP) augmentations were incorporated during the generation process, in the same spirit as other DSP- and synthesis-based audio augmentation strategies used to compensate for scarce training data [43, 44]. The metadata shows that to simulate different articulation patterns observed in flute performances, transitions (i.e., direct transitions, crossfades, and silence-separated note boundaries) were applied between the consecutive swaras. Also, multiple DSP enhancements, such as pitch shifting, time stretching, gain adjustment, additive noise, vibrato, and reverberation were added to specific notes or to segments of the melody. The augmentation for each note was noted in the metadata, so the whole synthesis process can be fully traced. These variations introduce the acoustic diversity in the dataset and enhance its ability to assess the robustness and generalization capability of the proposed framework of swara segmentation and classification. Figure 4 visualizes the usages of different DSP augmentation.

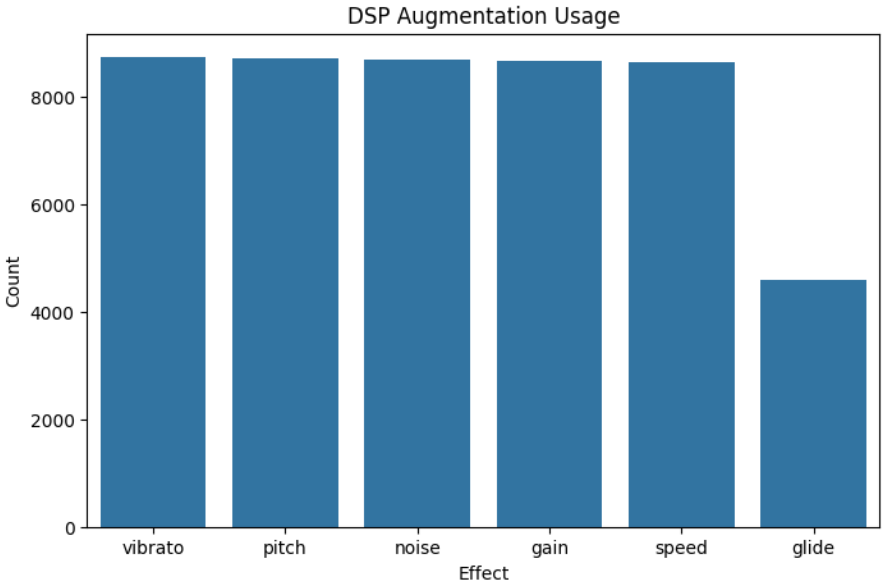


Figure 4: Count of different DSP augmentations used while creating Synthetic melody Dataset

The resulting metadata file can be used as a complete source of annotation for the synthetic dataset, so that direct calculation of statistics at the melody level, durations of the notes,

occurrence frequencies of the swaras, transition patterns between the notes, and the temporal distribution of the notes can be done. These annotations are detailed enough for quantitative segmentation accuracy assessment, as well as for swara classification performance assessment.

One of the major problems in automatic swara segmentation is the lack of accurate continuous musical recordings which are annotated. This is the reason for the emergence of the synthetic melody dataset. It is not easy to make a large-scale annotated dataset of flute performances in a short time and without much expertise, since all the note boundaries and swara labels need to be identified and verified one by one. A process like this is not only time consuming, but also hard to scale for various instruments and music styles. To address the aforementioned difficulty, a synthetic melody generation pipeline was created to automatically generate continuous melodies with precise ground-truth annotations without requiring much manual annotation.

The proposed framework was tested with recordings of the Indian classical flute, but the approach for generating the dataset is instrument agnostic, and can be modified for any monophonic instrument for which isolated note recordings are available. The proposed approach not only saves time in creating training data by automatically generating a continuous melody from single note samples, but also allows accurate time annotation. This allows for the creation of a strong swara segmentation and classification model, with no need to have hundreds of hours of manually labeled recordings, which would make it more scalable and applicable for further studies of monophonic musical instruments.

Chapter 4

METHODOLOGY

4.1 Overview

The chapter explains the proposed end-to-end approach of automating the swara transcription of continuous flute melodies of Indian classical music. The proposed system consists of two major stages namely note segmentation and swara classification. In the segmentation model, the individual notes in a continuous melody are separated by identifying the boundaries between them. The segmented notes are then fed to the classification model, which then predicts the corresponding swara for each note. Finally, the output of both models is merged to get the final transcribed swara of the input melody.

Our general pipeline consists of the following phases:

1. Data Collection
2. Synthetic Melody Data Creation
3. Segmentation model (CRNN + Attention)
    a. Extract Melody features
    b. Use synthetic data and extracted features to train the model
    c. Evaluate segmentation success in actual melodies
4. Classification Model (CNN + Attention)
    a. Extract Individual Swara features
    b. Use extracted features to train the classification model
    c. Evaluation
5. Merge Both Models
6. Upload test melodies and evaluate

As the two tasks are fundamentally different, this design was adopted. The segmentation model has to determine the time limits of single notes in a continuous melody while the classification model has to predict which swara the single note belongs to. The classification model was trained with a set of isolated swara recordings and thus needs each note segment as an input rather than a continuous swara melody. As a result, correct segmentation is one of the important prerequisites for proper classification of swaras. Moreover, the separation of the pipeline

allows for independent evaluation of both stages, and analysis of the segmentation and classification performance independently of each other, to facilitate the development, debugging and improvement of the model.

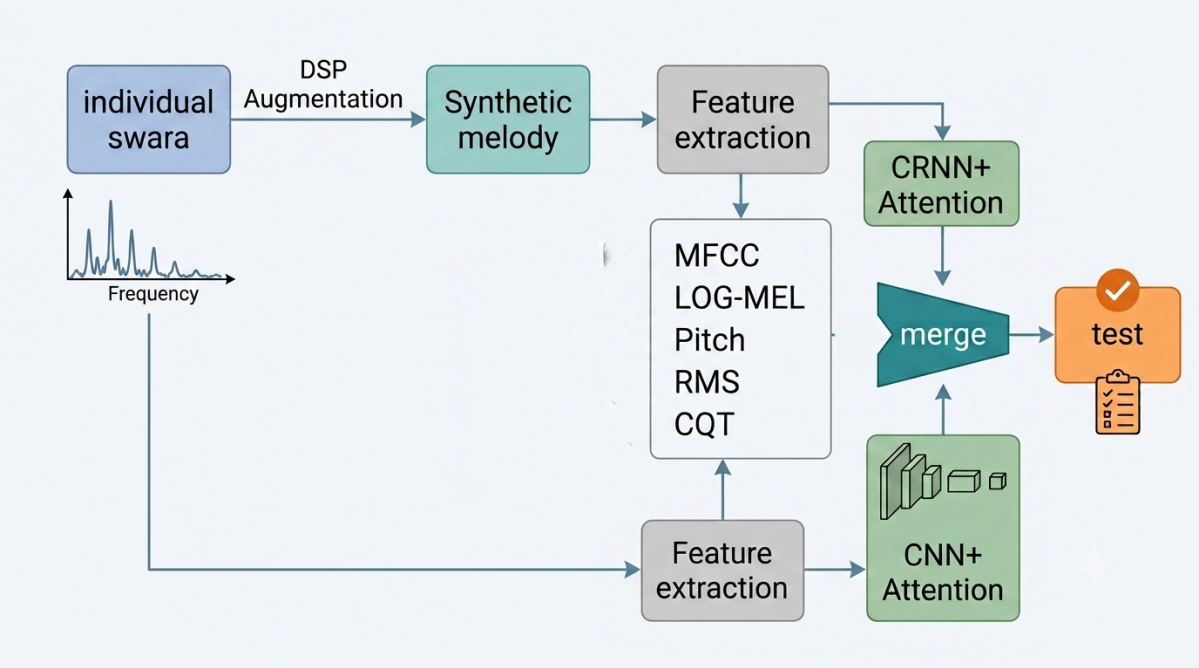


Figure 5: Methodology Overview

4.2 Overall Pipeline

4.2.1 Data collection

The initial phase of the proposed framework was the recording of individual swara notes of the Indian classical music scale, namely Sa, Re, Ga, Ma, Pa, Dha and Ni. The total number of swaras recorded was 748, from both publicly available recordings and self-recordings of flute playing. All the recordings were individually sorted into folders based on their swara class. The recordings were later resampled to 16kHz, mono-channel, .wav format.

The proposed pipeline is based on these stand-alone recordings. They have two main applications: (i) to create a synthetic melody set used for training the segmentation model and (ii) to train the swara classification model to detect individual notes. Chapter 3 describes in detail the data collection procedure, the data pre-processing, and the statistics of data sets.

Data Splitting. For classification model, the isolated swara dataset was split into 80-10-10 for training, testing, and validation and the similar was done with synthetic melodies for segmentation training. Feature extraction was conducted and was split according to train-test-validation split.

4.2.2 Feature Extraction

For both segmentation and classification training, we extracted various features from the audio files, such as

I. **MFCC:** A compact representation of the spectral characteristics of audio that is widely used in speech and audio recognition tasks [45].

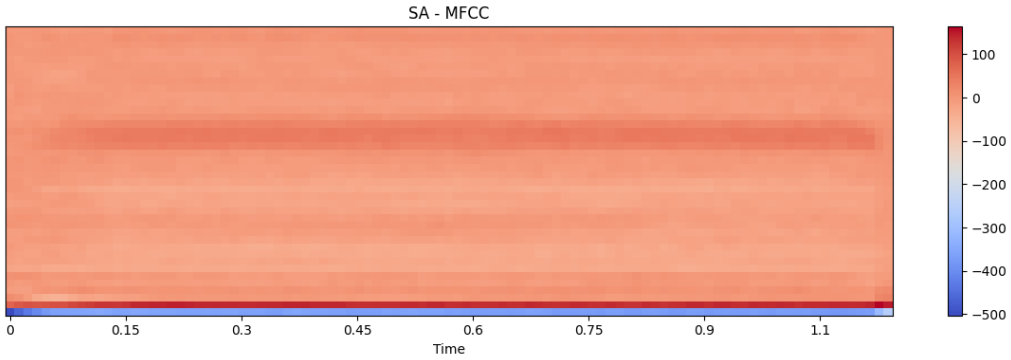


Figure 6: MFCC visualization of an individual note

II. **Log-Mel Spectrogram:** A logarithmically scaled Mel spectrogram that preserves more spectral information than MFCC while maintaining perceptual frequency scaling.

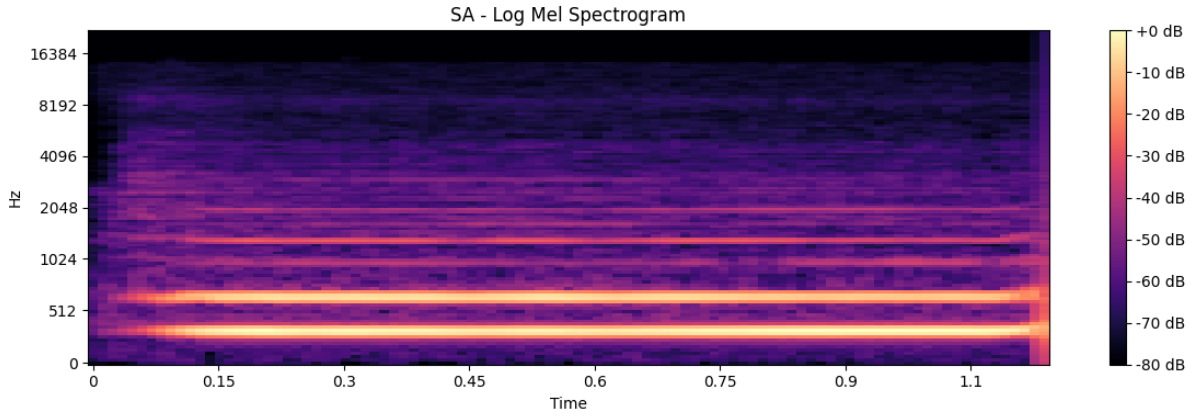


Figure 7: Log Mel spectrogram Visualization of an individual note

III. **Pitch:** Represents the fundamental frequency (F0) of the audio signal and provides direct information about musical note frequencies, estimated here using standard monophonic F0 estimation techniques [11, 13].

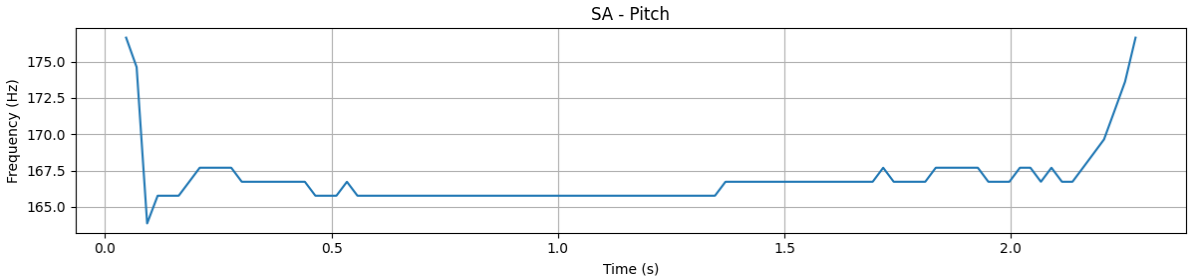


Figure 8: Pitch Visualization of an individual note

IV. **Root Mean Square (RMS) Energy:** Measures the short-term energy of the signal and assists in detecting note onsets and offsets.

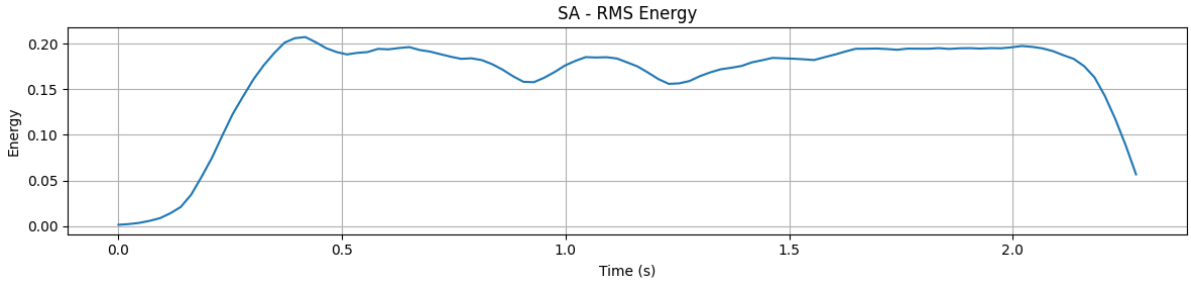


Figure 9: RMS energy visualization of an individual note

V. **Constant-Q Transformer (CQT):** A logarithmically spaced time-frequency representation specifically designed for musical signals [8]. Since musical notes are naturally distributed on a logarithmic frequency scale, CQT provides improved frequency resolution for note recognition and transcription tasks.

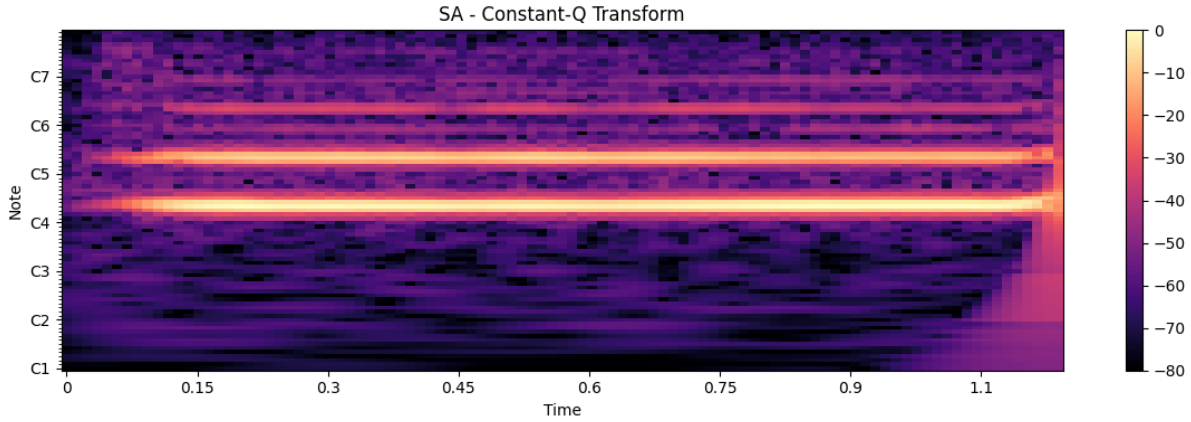


Figure 10: Constant-Q Transformation visualization of an individual note

Mel-frequency cepstral coefficients (MFCCs) are one of the most popular feature representations used in classification problems of speech and audio signals because of their compact representation of spectral characteristics. Early in this research MFCCs were used to extract features. MFCCs, however, are a compressed cepstral representation which discards a lot of detail spectral information from the original signal.

To ensure better time-frequency representation, the proposed scheme finally uses the log-Mel spectrogram as the main spectral representation. Log-Mel spectrograms preserve the two-dimensional spectral structure in a perceptually motivated manner, which is better for the CNN. A few pitch and RMS energy features were added to give complementary harmonic and energy information.

4.2.3 Experiments

To evaluate the effectiveness of different acoustic feature representations for automatic swara segmentation and classification, six experiments were conducted using the same overall processing pipeline. The same data set, processing steps, model architecture, training conditions and measures of effectiveness were used for all experiments. The only variable modified between

experiments was the input feature representation. This was a controlled experimental design, ensuring that any performance differences observed could only be attributed to the extracted features and not to differences in the model configuration or training process.

The experimental pipeline starts with the pre-processing of the audio input, and then goes on to extract the features using one of the selected acoustic representations. The extracted features then are given as input to the proposed deep learning architecture to be used for training and inferences. Finally, the trained models are tested with the same test set and the same performance criterion and thus a fair comparison of the effectiveness of each feature representation is possible.

To explore the suitability of the feature representations for the automatic segmentation and classification of swaras, the following feature representations were evaluated:

Experiment I: Log-Mel Spectrogram. Log-Mel spectrograms were chosen as a baseline spectral representation due to their ability to capture detailed information in time and frequency and approximate how humans hear frequencies. Unlike compressed cepstral features, Log-Mel spectrograms preserve the two dimensional spectral structure, which is very well suited for convolutional neural networks.

Experiment II: Pitch. The pitch information was also assessed independently to check if the fundamental frequency alone is sufficient to be used for distinguishing the individual swaras. Each swara is a particular pitch, so this experiment is designed to look at how well one can discriminate using pitch without additional spectral information.

Experiment III: RMS Energy. Root Mean Square (RMS) energy was assessed separately to investigate the contribution of the energy of the signal to segmentation and classification. RMS can be a potential cue for identifying note transitions between adjacent notes because there are usually energy changes at the note boundaries and places of articulation.

Experiment IV: Log-Mel Spectrogram + Pitch. This experiment is a mixture of spectral approach and explicit pitch estimation. The Log-Mel spectrogram shows the whole spectral shape of the audio signal, whereas the representation of pitch shows direct information about the fundamental frequency. The combination was expected to improve the discrimination between swaras with similar spectral characteristics.

Experiment V: Log-Mel Spectrogram + Pitch + RMS. The fifth experiment combines all the spectral, harmonic and energy information into a unified feature representation. Log-Mel spectrograms contain information about the spectral content, pitch captures the melodic information, and RMS addes temporal energy variations related to note transitions. The goal was to see if these complementary features would help to enhance segmentation and classification results.

Experiment VI: Constant-Q Transform (CQT). Evaluation of the Constant-Q Transform was made due to its logarithmically spaced frequency bins which closely resemble the

musical scale, offering increased frequency resolution at lower frequencies and increased temporal resolution at higher frequencies. CQT was designed to capture pitch-related characteristics better than conventional spectral representations, and was anticipated to provide a musically meaningful representation since swaras correspond to discrete musical notes.

Table 3: Experiment Table

Experiment	**Feature Representation**	**Motivation**
Experiment 1	Log-Mel	Baseline spectral representation
Experiment 2	Pitch	Harmonic Information Only
Experiment 3	RMS	Energy information only
Experiment 4	Log-Mel + Pitch	Spectral + harmonic features
Experiment 5	Log-Mel + Pitch + RMS	Multi-feature fusion
Experiment 6	CQT	Musically aligned frequency representation

4.2.4 Models

Segmentation Model. The proposed swara segmentation model is based on a Convolutional Recurrent Neural Network (CRNN) enhanced with an attention mechanism, following the same general architectural family of CNN feature extraction combined with recurrent and attention-based temporal modelling that has been used for onset and boundary detection in earlier works [23, 25, 28]. The model has been designed to identify the temporal transition between successive swaras of continuous flute melodies. Convolutional layers learn local spectral features within the input feature representation and recurrent layers learn long-term temporal dependencies between the audio sequence. Lastly, the attention mechanism highlights informative temporal regions to correctly detect note transition boundaries. The model was trained with the synthetic melody data introduced in Chapter 3, in which frame-level ground truth annotation was obtained automatically during melody synthesis.

The segmentation architecture and training configuration were kept the same for all feature extraction experiments (Experiments I-V) for a fair comparison across the various experiments. The only difference between experiments was the number of input channels (representing the feature representation selected). Single-feature experiments (Log-Mel, Pitch, and RMS) utilized one input channel, whereas feature fusion experiments employed two (Log-Mel + Pitch) and three (Log-Mel + Pitch + RMS) input channels, respectively. The CQT experiment used the same segmentation framework as the other experiments, but replaced the input representation with the CQT.

Table 4: Segmentation Model Architecture

Components	**Description**
Model	CRNN (Convolutional Recurrent Neural Network)
Input	Feature tensor of size $C \times 128 \times T$
Input Channels	1 (Log-Mel / Pitch / RMS), 2 (Log-Mel + Pitch), 3 (Log-Mel + Pitch + RMS), 1 (CQT)
Convolution Block	Two Conv2D blocks with Batch Normalization [46], ReLU activation, and Max Pooling
Attention Layer	Temporal attention for emphasizing important frames
Recurrent Layers	Bidirectional LSTM Layer [47, 48], Capture temporal dependencies between consecutive frames
Output Layer	Sigmoid activation producing boundary probabilities
Objective	Detect swara transition boundaries in continuous melodies

Table 5: Segmentation Training Configuration

Parameter	**Value**
Dataset	Synthetic Melody Dataset
Training/Validation Split	80%/20%
Random Seed	42
Batch Size	8
Number of Epochs	25
Optimizer	Adam
Initial Learning Rate	0.001
Learning Rate Scheduler	ReduceLROnPlateau
Scheduler Patience	3 Epochs
Scheduler Reduction Factor	0.5
Loss Function	Binary Cross-Entropy with Logits Loss (BCEWithLogitsLoss)
Positive Class Weight	15
Evaluation Metrics	Precision, Recall, F1-score

The boundary frames of the swara are much less than the total number of audio frames, making the segmentation task highly class-imbalanced. To overcome this, a positive class weight of 15 was used to increase the importance of the boundary frames during optimization, that is, Binary Cross-Entropy with Logits Loss was used. Due to the different length of melodies, all of the sequences were padded to a fixed length before batch training. In the process of computing the losses, a binary mask was used to remove padded frames from the loss computation process so that the model was trained only using input frames that are valid audio frames. The model was assessed by Precision, Recall, and F1-score to provide a thorough evaluation of the model's accuracy in detecting the boundaries.

Classification Model. The proposed swara classification model is based on a Convolutional Neural Network (CNN) enhanced with an attention mechanism. After segmentation, each segmented swara is transformed into the desired feature representation, and then given as input to the classifier. The convolutional layers extract discriminative spectral features from the input

representation, and the attention mechanism picks out the most informative parts of the feature maps, allowing the network to concentrate on the acoustical features that are most useful in distinguishing individual swaras. Last, the fully connected output layer predicts one of the seven swara classes (Sa, Re, Ga, Ma, Pa, Dha, Ni).

The classification architecture and training configuration were the same for all feature extraction experiments (Experiments I–V) to allow for a fair comparison. The only modification between experiments was the number of input channels corresponding to the selected feature representation. Single-feature experiments (Log-Mel, Pitch, and RMS) utilized one input channel, while feature fusion experiments employed two (Log-Mel + Pitch) and three (Log-Mel + Pitch + RMS) input channels, respectively. The input representation for the CQT experiment was the Constant-Q Transform, and the classification scheme remained the same.

Table 6: Classification Model Architecture

Component	**Description**
Model	CNN
Attention Layer	Spatial attention mechanism for emphasizing informative feature regions
Input	Feature tensor of size $C \times 64 \times 128$
Input Channels	1 (Log-Mel / Pitch / RMS), 2 (Log-Mel + Pitch), 3 (Log-Mel + Pitch + RMS), 1 (CQT)
Convolution Block	Three Conv2D blocks with Batch Normalization [46], ReLU activation, and Max Pooling
Adaptive Pooling	Adaptive Average Pooling to aggregate spectral information while preserving temporal resolution
Fully Connected Layer	Maps extracted features to swara classes
Output Layer	Fully connected layer producing probabilities for seven swara classes
Objective	Classify segmented notes into one of the seven swara classes

Table 7: Classification Training Configuration

Parameter	**Value**
Dataset	Individual Swara Dataset
Training/ Validation Split	80% / 20%
Random Seed	42
Batch Size	32
Number of Epochs	25
Optimizer	Adam
Initial Learning Rate	0.001
Learning Rate Scheduler	ReduceLROnPlateau
Scheduler Patience	3 Epochs
Scheduler Reduction Factor	0.5
Loss Function	Cross-Entropy Loss
Evaluation Metrics	Accuracy, Precision, Recall, F1-score

The classification model was trained with the dataset of individual swara as described in

Chapter 3. All the feature tensors were padded or cropped to a fixed temporal dimension for uniform batch training input dimensions. The optimal network parameters were obtained with Adam optimizer [49] and a learning rate of 0.001. To encourage stable convergence during training, a ReduceLROnPlateau scheduler was used which automatically reduced the learning rate whenever the validation loss levelled off. The same validation protocol was used for all experiments, so that any difference in classification performance was due to the choice of feature representation.

Attention Mechanism. Incorporating an attention mechanism [6] into a CRNN for audio segmentation is a huge improvement because it allows the network to dynamically focus on the most relevant part of any sequence instead of treating all time frames the same. Different parts of an audio stream have very different levels of significance, e.g., a longer recording with overlapping sounds or changing acoustical settings. Recurrent networks, such as LSTMs, take in inputs one by one, but may have difficulty holding the long-range context of longer audio segments. Attention mechanisms overcome this by calculating weighted importance scores for all of the time steps at once, so that the model could "attend" directly to important transition boundaries or event markers (such as the precise beginning or end of a spoken word or environment noise). This significantly enhances temporal accuracy in audio segmentation.

For standalone CNN classifiers in audio classification, attention mechanisms (typically represented as spatial or channel-wise attention blocks, e.g., Squeeze-and-Excitation networks [50]) boost performance by selectively emphasizing the most informative regions of a Mel-spectrogram. The CNN extracts local texture and frequency features, but not all the frequency bands or the harmonic patterns equally contribute to the identification of a sound class. By attending to discriminative acoustic cues, such as a particular formant in voice recognition or a particular high-frequency snap in soft percussion, the model can selectively enhance the relevant feature maps while suppressing irrelevant acoustic features and background noise. These feature channels and spatial regions are selectively weighted so that the CNN is far more powerful and effective, even with noisy or complex audio information.

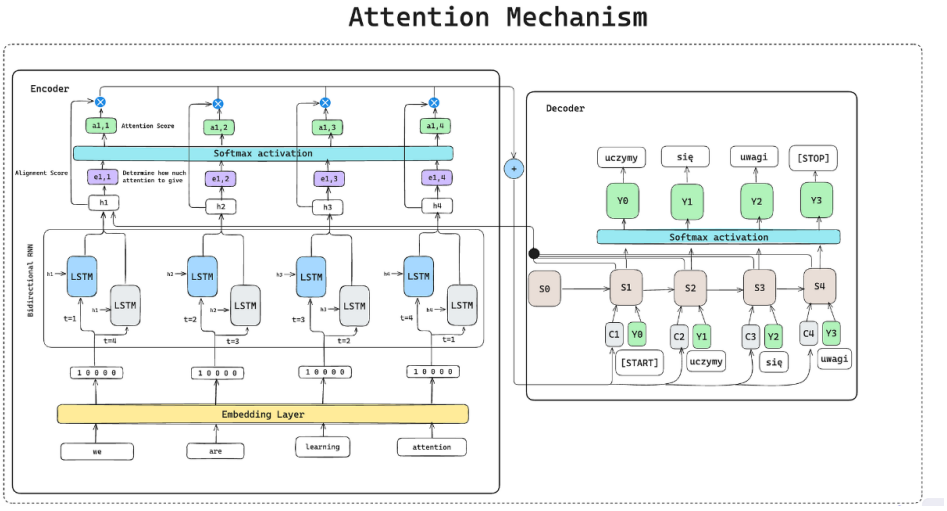


Figure 11: Architectural diagram of Attention Mechanism

4.2.5 Training Configurations

All the experiments were carried out using Python and the PyTorch deep learning framework [51] and performed on the Google Colab platform. To effectively process the segmentation and classification models, model training and evaluation were sped up with an NVIDIA Tesla T4 Graphics Processing Unit (GPU). To make a fair comparison between different feature representations in the different experiments, identical training configurations were used. The software and hardware used in this work are summarized in Table 8 for details.

Table 8: Experimental Environment

Component	**Specification**
Programming Language	Python 3
Deep Learning Framework	PyTorch
Audio Processing Library	Librosa [52]
Numerical Computing	NumPy
Data Manipulation	Pandas
Visualization	Matplotlib
Development Environment	Google Colab
Hardware Accelerator	NVIDIA Tesla T4 GPU
Operating Environment	Ubuntu(Google Colab Runtime)

4.2.6 Evaluation Metrics

The performance of the proposed framework was evaluated using several widely adopted classification and segmentation metrics, including Accuracy, Precision, Recall, and F1-score. These metrics provide complementary insights into the effectiveness of the proposed models by measuring both the correctness of predictions and the ability to accurately detect swara boundaries.

Accuracy: It measures the overall proportion of correctly predicted samples among all predictions and is defined as

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{4.1}$$

where TP, TN, FP, and FN denote True Positives, True Negatives, False Positives, and False Negatives, respectively.

Precision: It evaluates the proportion of predicted positive samples that are actually correct. It measures how reliable the positive predictions of the model are and is expressed as

$$\text{Precision} = \frac{TP}{TP + FP} \tag{4.2}$$

Recall: It measures the ability of the model to correctly identify all positive samples and is calculated as

$$\text{Recall} = \frac{TP}{TP + FN} \tag{4.3}$$

The **F1-score** is the harmonic mean of Precision and Recall, providing a balanced evaluation when both false positives and false negatives are important. It is computed as

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{4.4}$$

The F1-score is particularly suitable for imbalanced datasets, where relying solely on Accuracy may produce misleading conclusions.

4.2.6.1 Evaluation Protocol

Different evaluation metrics were employed for the segmentation and classification tasks according to their respective objectives.

In the classification model, the accuracy and F1 score were used as evaluation metrics. The accuracy value is a measure of the overall correctness of the swara prediction and the F1-score is a balanced measurement of the classification performance, considering both precision and recall, which makes it more suitable when the class distribution is not perfectly even.

In the segmentation model, the detection of swara boundaries was evaluated based on Precision, Recall, and F1-score. Precision is defined as how many of the predicted boundaries correctly correspond to actual swara transitions; Recall is defined as how many of the actual swara transitions are correctly detected; and the F1 score is a balanced measure of segmentation performance, combining precision and Recall.

Chapter 5

RESULT AND DISCUSSION

To evaluate the effectiveness of the proposed framework, six experiments were conducted using different acoustic feature representations while maintaining identical model architectures and training configurations. This ensured that any variation in performance could be attributed solely to the selected feature representation rather than differences in model design or optimization.

Based on the literature reviewed in Chapter 2, Log-Mel spectrograms have been widely adopted for audio and music processing tasks because of their ability to represent both spectral and perceptual characteristics of sound. Therefore, it was initially hypothesized that Log-Mel would achieve the best overall performance for both swara segmentation and classification. In addition to Log-Mel, Pitch, RMS, Log-Mel + Pitch, Log-Mel + Pitch + RMS, and Constant-Q Transform (CQT) representations were investigated. While all feature representations produced satisfactory results to varying degrees, the CQT-based model achieved the highest overall performance across the proposed end-to-end framework.

5.1 Experiment 1: Log-Mel

In the first experiment, the input feature representation for segmentation and classification were both a Log-Mel spectrogram. Based on previous studies in music information retrieval and audio classification, Log-Mel was initially expected to provide the highest overall performance due to its ability to capture both spectral and perceptual characteristics of audio signals.

For swara classification, the CNN with Attention model was trained for 30 epochs using the Log-Mel feature representation. The model consisted of 101,960 trainable parameters and achieved an overall classification accuracy of 85.0% with an F1-score of 89.3%. The training and validation loss and accuracy curves are presented in Figure 12. Both training and validation curves are smooth, suggesting stable learning without substantial overfitting and that Log-Mel provides an adequate amount of spectral information for accurate swara classification.

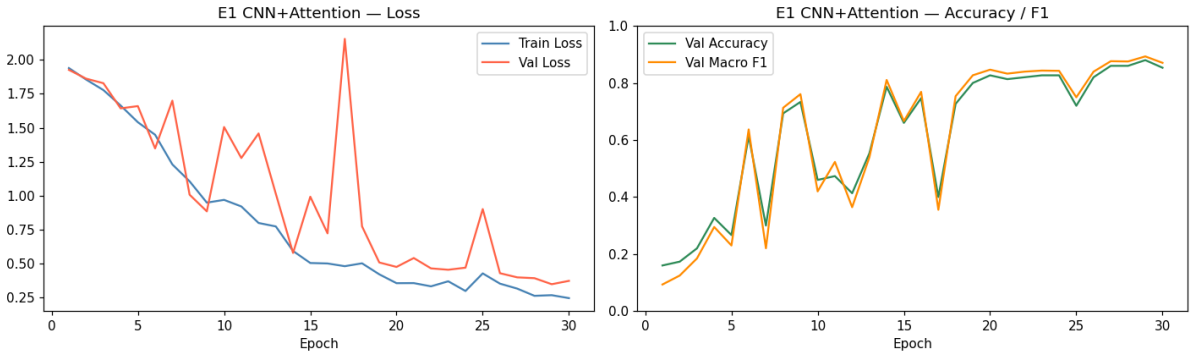


Figure 12: Training and validation loss, accuracy, and F1-score of the classification model for Experiment 1.

For swara segmentation, the CRNN with Attention model was trained for 25 epochs using the same Log-Mel feature representation. There were 1,090,594 trainable parameters in the segmentation network. Precision, Recall and F1-score were used to evaluate the model for boundary detection. The training curves for the segmentation model are presented in Figure 13 and display the evolution of the segmentation model during training.

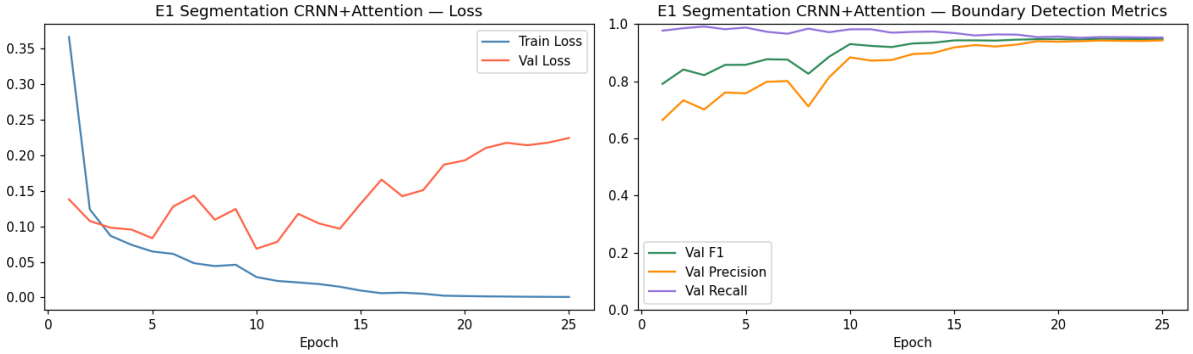


Figure 13: Training loss and boundary detection metrics (Precision, Recall, and F1-score) of the segmentation model for Experiment 1.

5.1.1 End to end pipeline for experiment 1

The segmentation and classification models were trained independently and then combined to assess the entire swara identification process. Every test melody was processed through the segmentation stage followed by the classification stage using the trained Log-Mel models.

The segmentation was evaluated from the percentages of the melodies that were perfectly segmented, that is, the boundaries of all swaras were correctly identified. Only these perfectly segmented melodies were used in the classification evaluation to prevent any classification error due to segmentation errors from influencing the reported classification accuracy. Using this evaluation protocol, the proposed framework achieved a segmentation accuracy of 71.43%, while the classification stage achieved a note-level accuracy of 84.29% on the correctly segmented melodies. The following is an example of the full pipeline output created by the proposed pipeline, Figure 14.

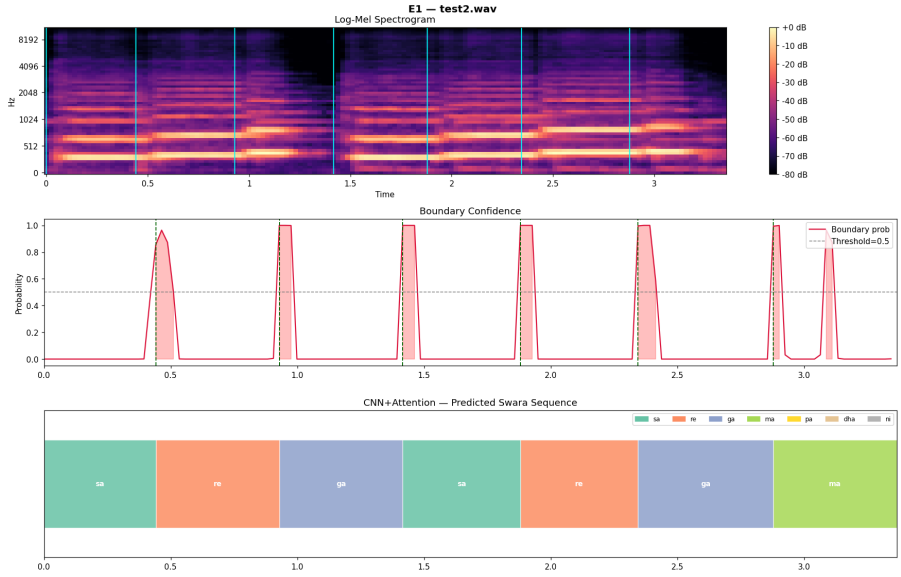


Figure 14: End to end pipeline visualization of experiment 1

5.2 Experiment 2: Pitch

The second experiment investigated the effectiveness of Pitch as the sole feature representation for both segmentation and classification. Pitch contains far less spectral information compared to Log-Mel, as it represents only the fundamental frequency of the audio signal. The present experiment was undertaken to test if pitch information alone is enough for the accurate identification of swaras.

The CNN with Attention classifier was trained using the same architecture and training configuration as Experiment 1, with Pitch replacing Log-Mel as the input feature. The model achieved an overall classification accuracy of 69.0% and an F1-score of 71.5%. However, a significant drop in performance was noted when compared to Experiment 1 suggesting that while Pitch is effective at capturing note frequency, it does not contain the harmonic and timbral information required for good discrimination of acoustically similar swaras.

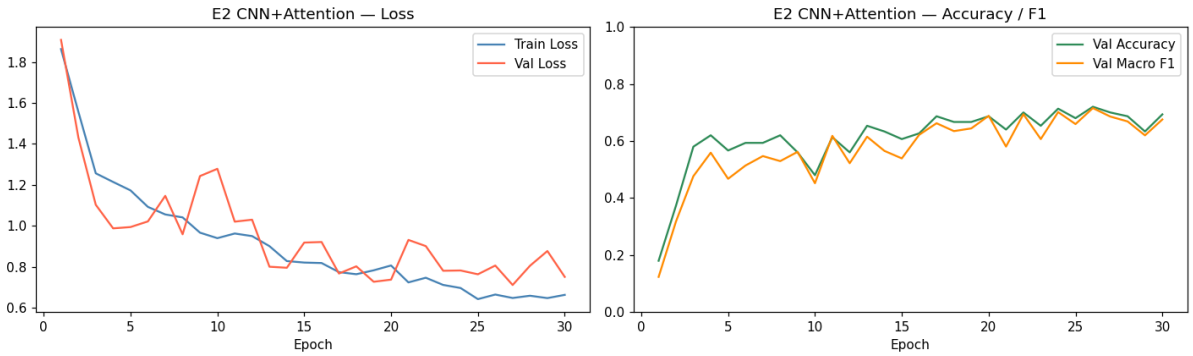


Figure 15: Training and validation loss, accuracy, and F1-score of the classification model for Experiment 2.

For segmentation, the CRNN with Attention model followed the identical architecture and training configuration used in Experiment 1. The model achieved a Precision of 47.0%, Recall

of 93.4%, and an F1-score of 57.7%. The high Recall value means that the model correctly identified the majority of the true swara boundaries while the comparatively low Precision value means that the model resulted in many false swara boundary predictions, giving a lower overall F1-score.

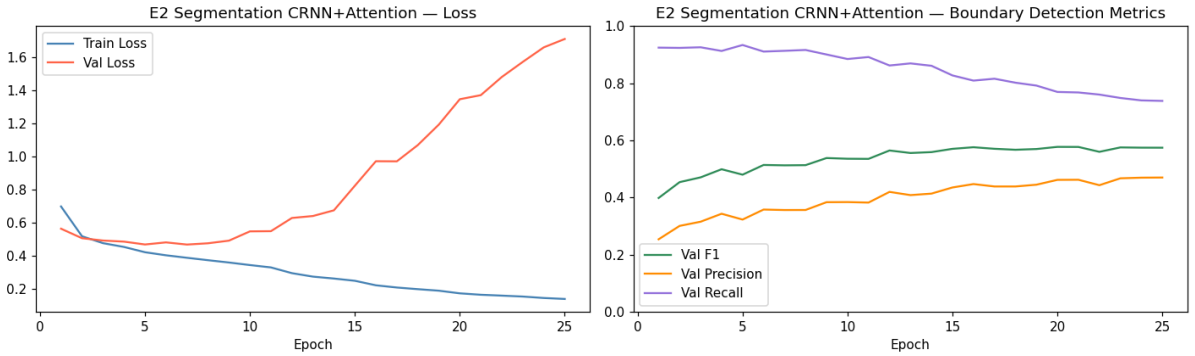


Figure 16: Training loss and boundary detection metrics (Precision, Recall, and F1-score) of the segmentation model for Experiment 2.

5.2.1 End to End Pipeline for Experiment 2

The complete end-to-end pipeline was evaluated following the same procedure as Experiment 1. All the test melodic pieces were passed through both the segmentation and classification stages with Pitch as the input feature representation. The proposed system has been evaluated according to the aforementioned evaluation protocol, and the accuracy of segmentation was obtained as 40.00%. The classification model had a note level accuracy of 21.43%, among perfectly segmented melodies. This is because there is a significant difference in end-to-end performance as compared to Experiment 1, which illustrates that Pitch alone is not enough to develop a reliable swara identification system, especially for segmentation, where accurate boundary detection depends on reliable spectral information.

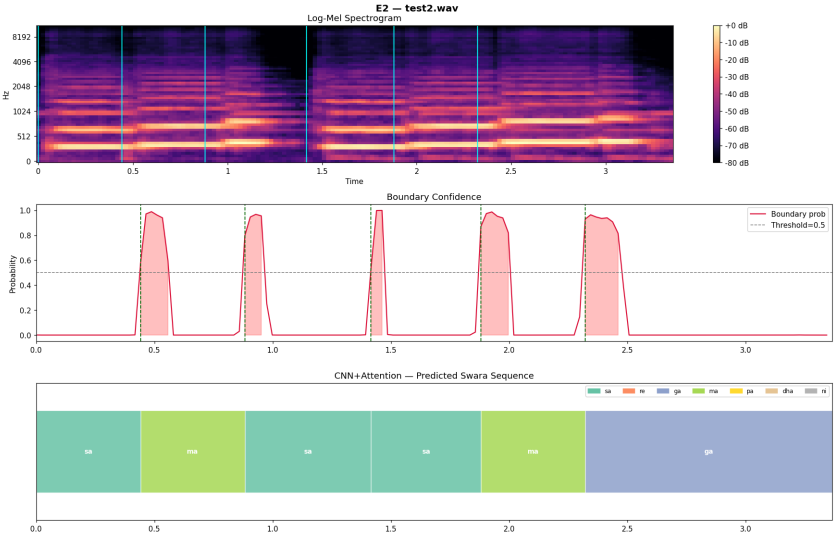


Figure 17: End to end pipeline visualization of experiment 2

5.3 Experiment 3: RMS

The third experiment investigated the effectiveness of Root Mean Square (RMS) Energy as the sole feature representation for both swara segmentation and classification. Unlike Log-Mel and Pitch, RMS represents only the energy or loudness variations of the audio signal and does not contain any spectral or frequency information. This experiment was carried out to check whether energy is enough to give sufficient information to identify swaras.

For swara classification, the CNN with Attention model was trained using the RMS feature representation while maintaining the same architecture and training configuration as the previous experiments. The model achieved an overall classification accuracy of 31.0% with an F1-score of 29.6%. The relatively poor performance suggests that RMS energy is not enough by itself to differentiate between different swaras as there can be cases where two or more notes can have similar energy patterns, but different spectral characteristics.

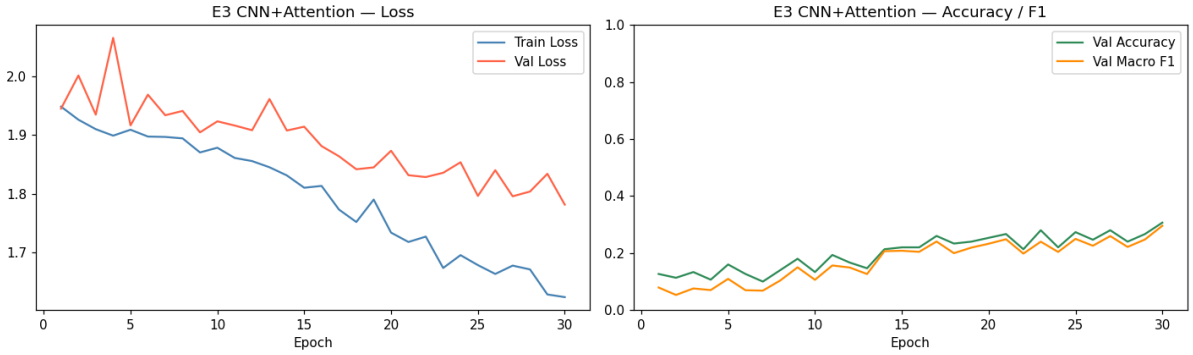


Figure 18: Training and validation loss, accuracy, and F1-score of the classification model for Experiment 3.

For swara segmentation, the CRNN with Attention model achieved a Precision of 61.9%, Recall of 97.2%, and an F1-score of 74.8%. Although RMS lacks frequency information, the model was still able to identify a large proportion of swara boundaries, as reflected by the high Recall. However, the comparatively lower Precision suggests that the model generated a considerable number of false boundary detections.

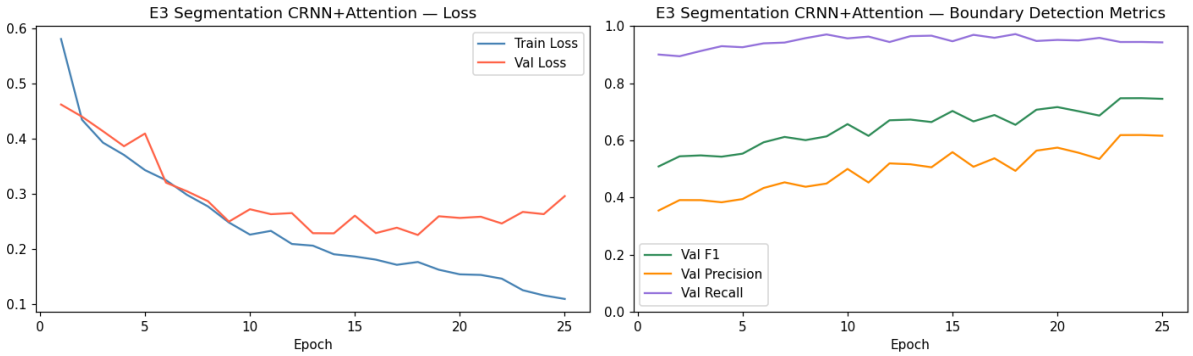


Figure 19: Training loss and boundary detection metrics (Precision, Recall, and F1-score) of the segmentation model for Experiment 3.

5.3.1 End-to-End Pipeline

The complete end-to-end pipeline was evaluated following the same procedure used in the previous experiments. For all the test melodies, both segmentation and classification were applied using RMS as the representation of the input. The proposed framework achieved a segmentation accuracy of 64.29%. Among the perfectly segmented melodies, the classification model achieved a note-level accuracy of 14.29%. The segmentation performance with the RMS method was moderate, but due to the absence of discriminative spectral information, the classification performance was poor, making it unsuitable as a standalone feature for automatic swara identification.

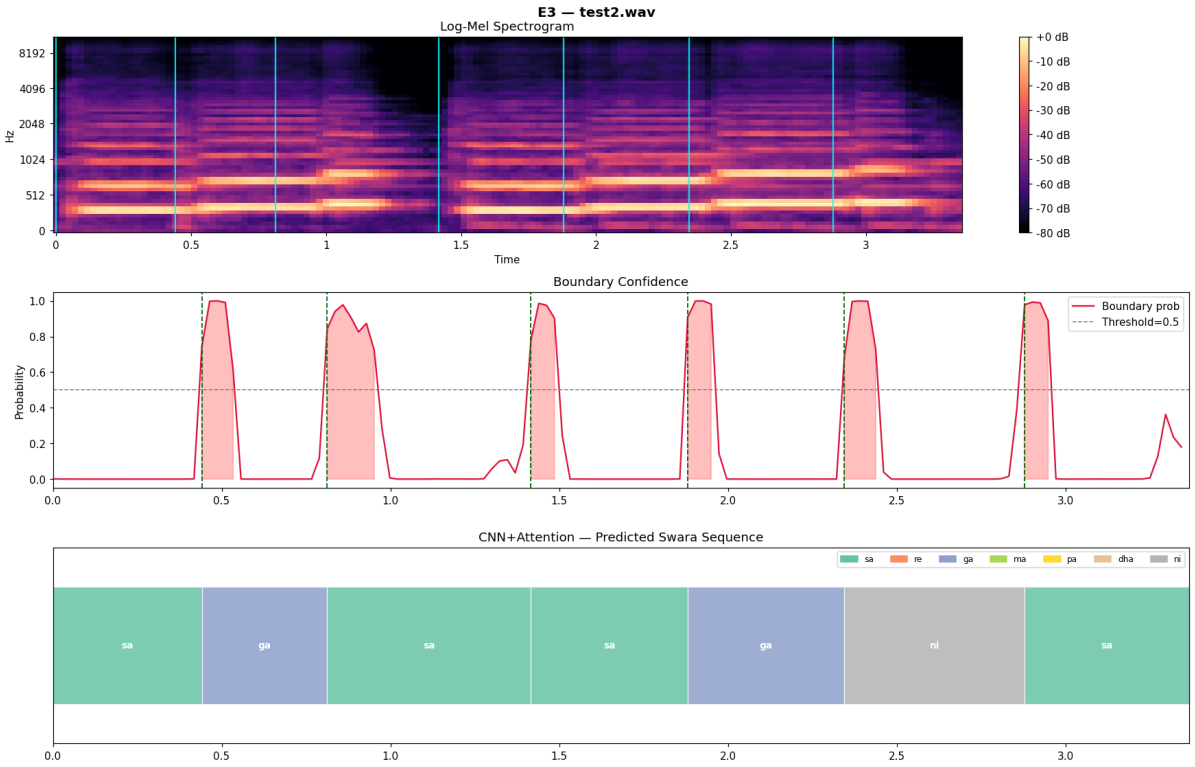


Figure 20: Example visualization of the end-to-end swara segmentation and classification pipeline for Experiment 3.

5.4 Experiment 4: Log-Mel + Pitch

The fourth experiment combined Log-Mel spectrograms and Pitch to investigate whether combining spectral and frequency information could improve segmentation and classification performance. The same model architectures and training configurations were maintained, with the only difference being the use of two input channels corresponding to the fused feature representation.

For swara classification, the CNN with Attention model achieved an overall classification accuracy of 84.7% with an F1-score of 86.5%. The overall representation achieved a comparable performance to the Log-Mel-only representation, that is, there was very little added gain in the classification performance with the addition of Pitch information.

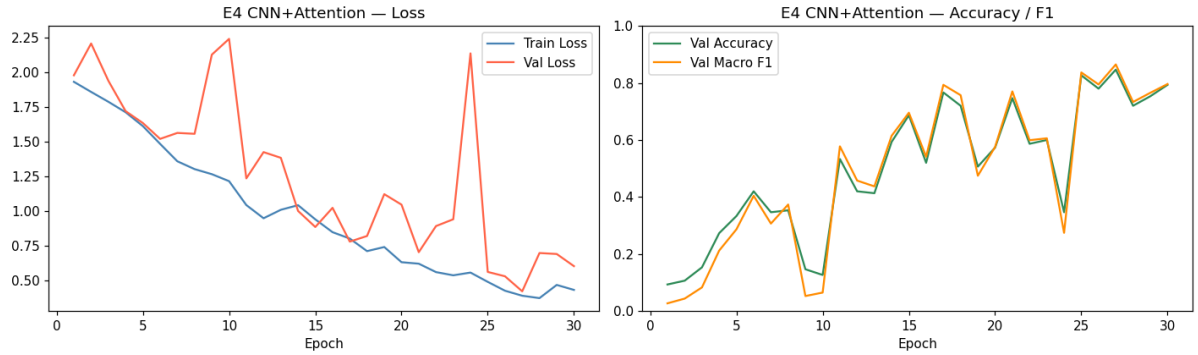


Figure 21: Training and validation loss, accuracy, and F1-score of the classification model for Experiment 4.

For swara segmentation, the CRNN with Attention model achieved a Precision of 94.3%, Recall of 98.7%, and an F1-score of 94.9%. These results are a significant improvement over the previous experiments which showed that the spectral and pitch information can be used to identify swara transition boundaries with much greater reliability.

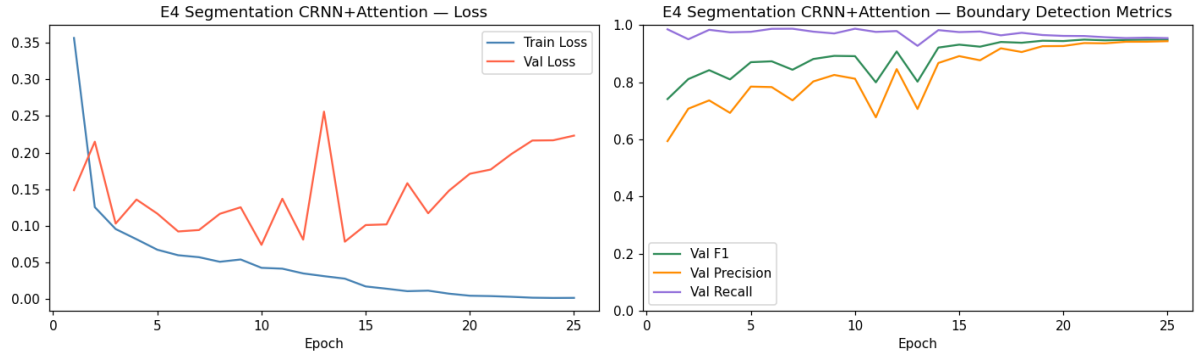


Figure 22: Training loss and boundary detection metrics (Precision, Recall, and F1-score) of the segmentation model for Experiment 4.

5.4.1 End-to-End Pipeline Evaluation

The Log-Mel and Pitch representation were used together to evaluate the entire pipeline. The proposed framework achieved a segmentation accuracy of 57.14%, while the classification stage achieved a note-level accuracy of 39.29% on the correctly segmented melodies. Although the segmentation model performed well during the validation process at frame level, the end-to-end evaluation illustrates that perfectly segmenting an entire melody is still a difficult problem, leading to sub-optimal overall pipeline performance.

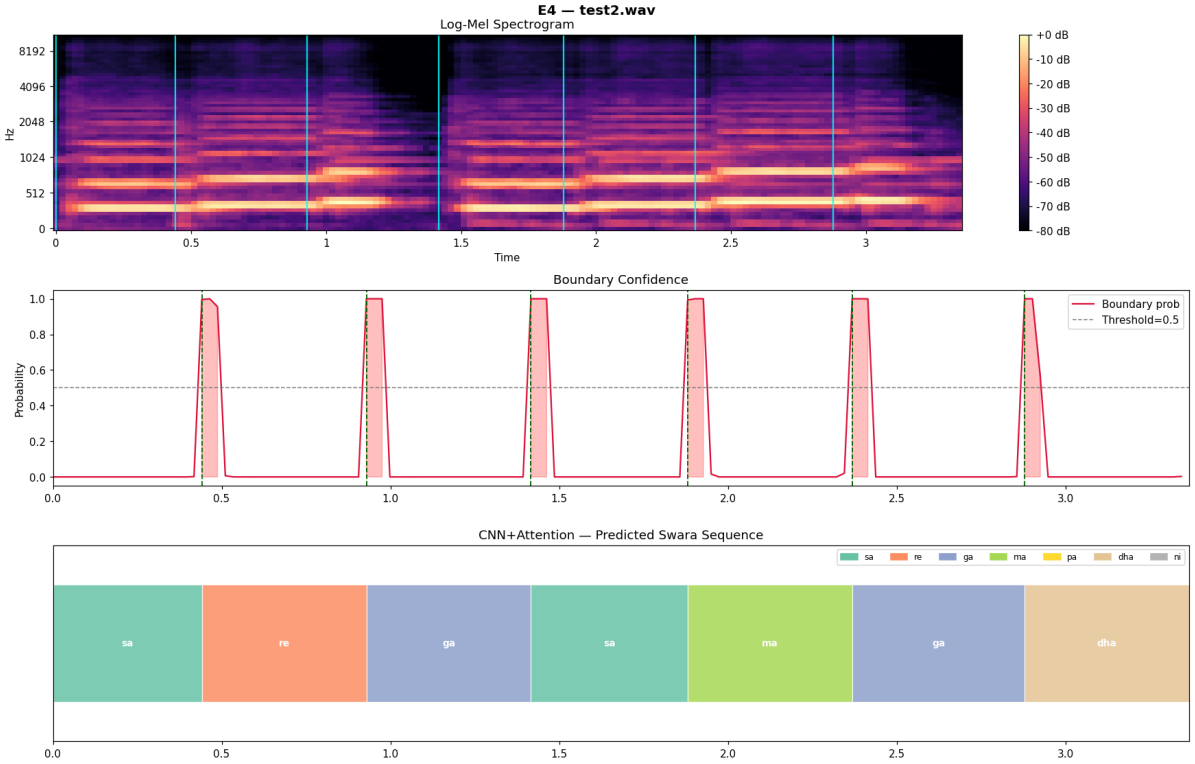


Figure 23: Example visualization of the end-to-end swara segmentation and classification pipeline for Experiment 4.

5.5 Experiment 5: Log-Mel + Pitch + RMS

In the fifth experiment, Log-Mel, Pitch, and RMS were used together as a three-channel input. The objective of this experiment was to investigate whether the combination of energy features with spectral and pitch features could enhance the performance of the proposed framework further.

For swara classification, the CNN with Attention model achieved an overall classification accuracy of 93.3% with an F1-score of 94.0%, representing a significant improvement over the previous experiments. The combination of complementary acoustic information enabled the classifier to learn more discriminative feature representations, resulting in the highest classification performance observed thus far.

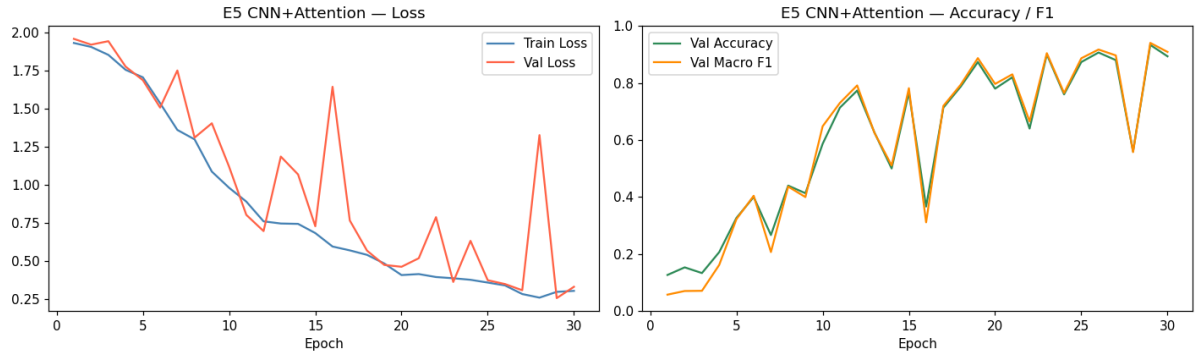


Figure 24: Training and validation loss, accuracy, and F1-score of the classification model for Experiment 5.

For swara segmentation, the CRNN with Attention model achieved a Precision of 93.5%, Recall of 99.3%, and an F1-score of 94.4%. Similar to Experiment 4, the segmentation model demonstrated excellent boundary detection performance, with both Precision and Recall remaining consistently high throughout training.

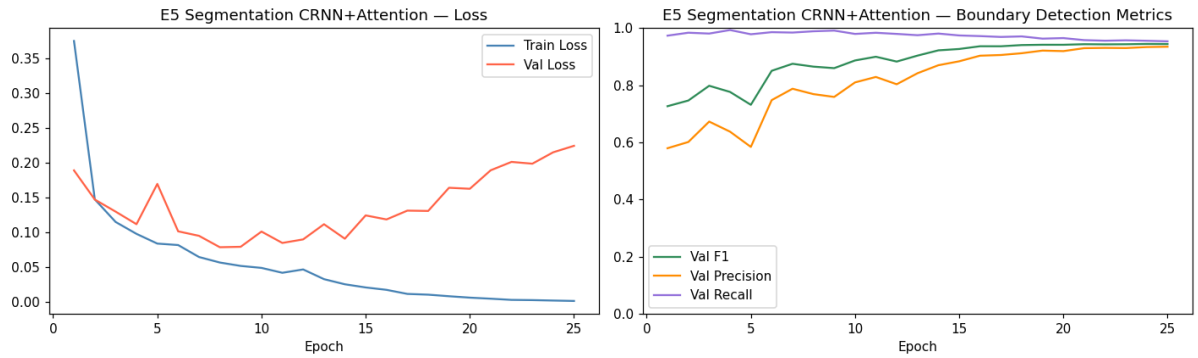


Figure 25: Training loss and boundary detection metrics (Precision, Recall, and F1-score) of the segmentation model for Experiment 5.

5.5.1 End-to-End Pipeline Evaluation

The complete end-to-end pipeline was evaluated using the three-channel feature representation. The proposed framework achieved a segmentation accuracy of 50.00%. Among the perfectly segmented melodies, the classification model achieved a note-level accuracy of 75.51%. The substantial improvement in classification accuracy demonstrates the effectiveness of combining complementary acoustic features. However, the lower melody-level segmentation accuracy suggests that accurately detecting boundaries throughout an entire melody remains the main factor limiting the overall performance of the complete pipeline.

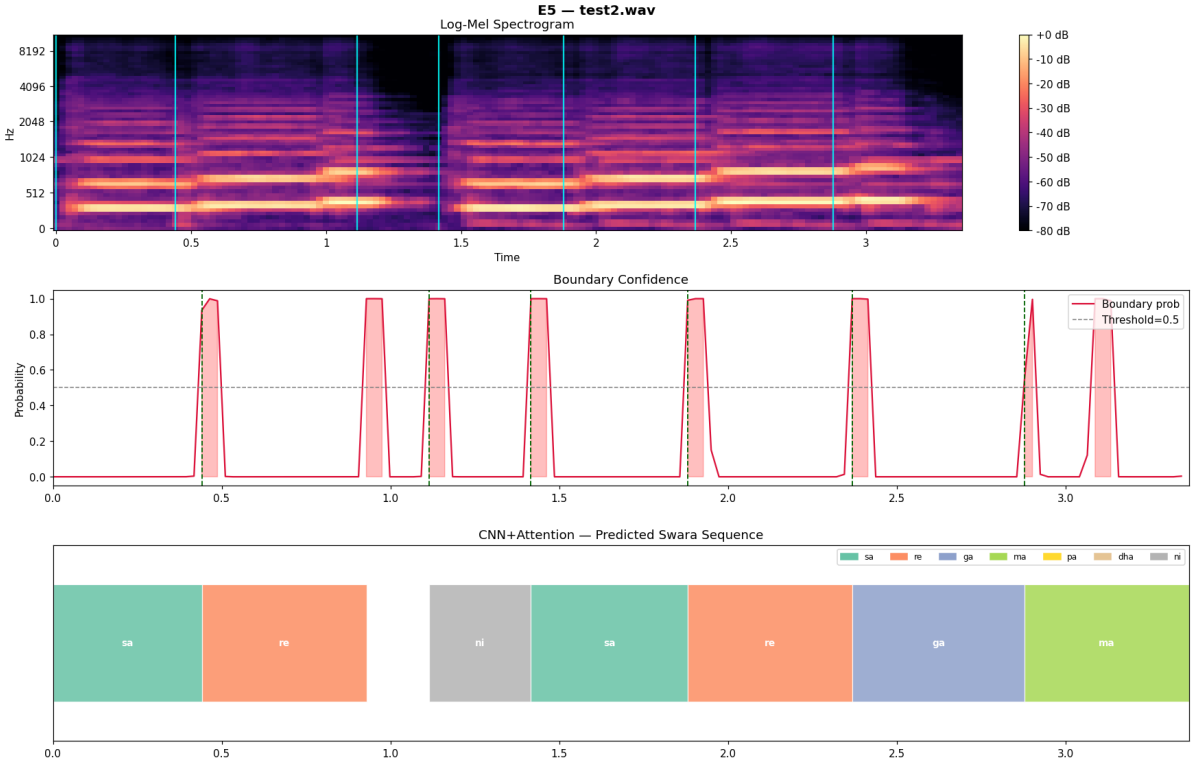


Figure 26: Example visualization of the end-to-end swara segmentation and classification pipeline for Experiment 5.

5.6 Experiment 6: Constant-Q Transform (CQT)

In the sixth experiment, the Constant-Q Transform (CQT) was used as input feature representation for both segmentation and classification models. CQT uses logarithmically spaced frequency bins which are more closely related to the musical scale than the previous feature representations, and are well suited to music signal analysis and monophonic instrument recognition. Because the flute performances are made up of isolated musical notes and have very clear harmonic frameworks, a spectrogram-like feature was not seen as informative as CQT.

For swara classification, the CNN with Attention model was trained for 30 epochs using the same network architecture and training configuration employed in the previous experiments. Throughout training, both the training and validation accuracies increased consistently, indicating stable convergence of the model. The best validation accuracy of 99.33% was achieved at Epoch 26, after which the performance remained stable until the completion of training. These results show that the CQT representation provides a lot of discriminative information for distinguishing between the seven swara classes.

For swara segmentation, the CRNN with Attention model was trained for 20 epochs using the CQT feature representation. The training loss decreased steadily from 0.0704 during the first epoch to 0.0257 at the final epoch, indicating stable optimization and convergence throughout the training process. The loss tends to be progressively reduced, indicating that the model was able to capture the temporal characteristics needed for the detection of swara transition boundaries in continuous flute melodies.

5.6.1 End-to-End Pipeline Evaluation

The complete end-to-end pipeline was evaluated following the same procedure adopted for the previous experiments. All test melodies have been passed through the segmentation and classification stages using the trained CQT models. Segmentation performance was measured based on the percentage of melodies that were segmented perfectly, while classification accuracy was calculated only for the correctly segmented melodies.

The proposed framework achieved the highest overall performance among all six experiments, obtaining a segmentation accuracy of 92.86%. Furthermore, the classification model correctly identified 97.80% of the swaras within the perfectly segmented melodies. These results demonstrate that the Constant-Q Transform provides the most suitable acoustic representation for the proposed end-to-end swara segmentation and classification framework, outperforming all other investigated feature representations.

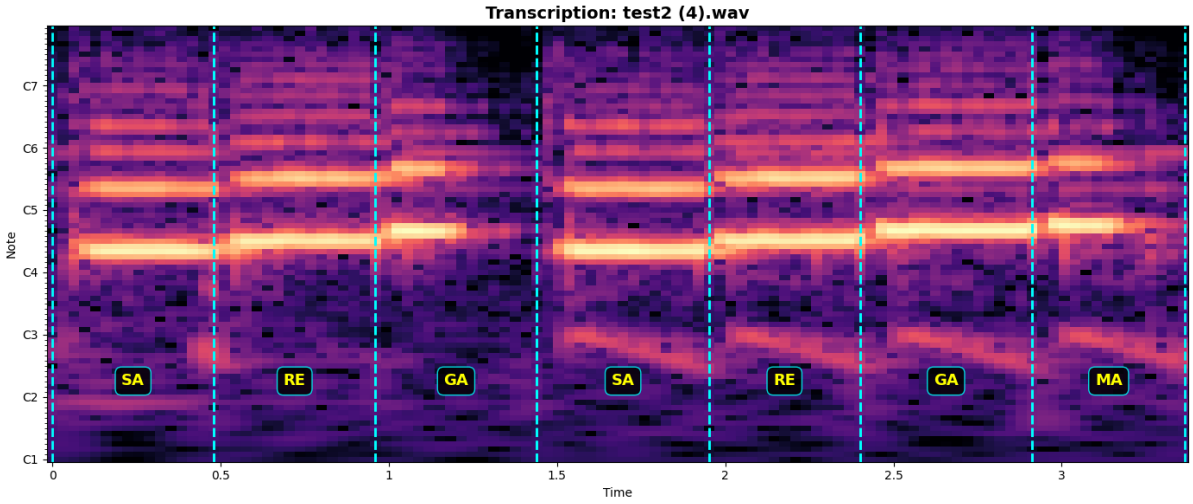


Figure 27: Example visualization of the end-to-end swara segmentation and classification pipeline for Experiment 6.

Table 9: Overall Performance Comparison of All Experiment

Experiment	Feature	Classification Accuracy (%)	Classification F1 (%)	Segmentation Precision (%)	Segmentation Recall	Segmentation F1	End-to end Segmentation	End to end Classification
E1	Log-Mel	85.00	89.30	94.3	98.9	95.0	71.43	84.29
E2	Pitch	69.00	71.50	47.00	93.40	57.70	40.00	21.43
E3	RMS	31.00	29.60	61.90	97.20	74.80	64.29	14.29
E4	Log-Mel + Pitch	84.70	86.50	94.30	98.70	94.90	57.14	39.29
E5	Log-Mel + Pitch + RMS	93.30	94.00	93.50	99.30	94.40	50.00	75.51
E6	CQT	99.33	99.00	100.00	98.98	99.49	92.86	97.80

5.7 Discussion

The experimental results demonstrate that the choice of acoustic feature representation has a significant impact on both swara segmentation and classification performance. While the six feature representations investigated were all based on the same model architectures, training method, optimization approach, and evaluation protocol, significant performance variability can be observed. The differences here suggest that the quality and type of acoustic features extracted are a more important determinant than the model architecture.

The overall performance of the Constant-Q Transform (CQT) was the best out of the six experiments. The end-to-end pipeline achieved a segmentation accuracy of 92.86% and a note-level classification accuracy of 97.80% which outperformed all other feature representations tested in this work. CQT is different from standard spectrogram-based features, as it uses a logarithmic scale for frequency, which is close to the spacing of musical notes. Since Indian classical flute performances are inherently musical and consist of discrete swaras, this representation preserves harmonic relationships more effectively, enabling both the segmentation and classification models to learn more discriminative musical patterns.

The Log-Mel spectrogram also performed well throughout all experiments. As one of the most widely used feature representations in speech and music processing, Log-Mel effectively captures both spectral and perceptual characteristics of audio signals. The experimental results show that Log-Mel is a reliable representation for automatic swara identification, and yields good segmentation and classification results. However, despite its strong results, it remained inferior to the CQT representation, suggesting that feature representations specifically designed for musical signals are more suitable for this application.

The results of the experiments using Pitch and RMS alone were significantly poorer. In pitch, only the fundamental frequency of the audio signal is used and the other harmonic content and timbral information are removed, making it hard for the models to tell the difference between acoustically similar swaras. RMS was even worse for classification, due to the fact that energy information alone contains little discriminative information about musical note identity. Interestingly, the model had a relatively high Recall during the segmentation, but a relatively low Precision, meaning that it identified many correct note boundaries, but also made many false predictions for note boundaries. From these observations, it is found that neither Pitch nor RMS alone provides sufficient information for robust end-to-end swara identification.

The different acoustic representations are complementary as demonstrated by the feature fusion experiments. The combination of Log-Mel and Pitch significantly enhanced the segmentation measures, showing that the two types of information together offer a more complete representation of note transitions. Furthermore, incorporating RMS into the feature set resulted in the highest classification performance among the non-CQT experiments. These results indicate that the use of RMS alone is not enough to make a correct classification, but it is a useful complement to more informative spectral features. The results demonstrate that feature fusion enables the models to exploit multiple acoustic characteristics simultaneously, leading to improved overall performance.

An interesting observation can be made by comparing the segmentation evaluation metrics with the end-to-end segmentation accuracy. The frame-level F1-scores of several experiments are above 94%, but the accuracy of the melody-level segmentation is still significantly lower. This difference is due to the evaluation protocol used in this study. The frame-level Precision, Recall and F1-Score assess the correctness of the boundary predictions for each frame, while the end-to-end segmentation score considers a melody to be correctly segmented when all entries in the boundary prediction for the melody are correct. Thus, even if one segment is incorrectly identified as a boundary, the entire melody is wrongfully regarded as incorrectly segmented. This assessment criterion is much more demanding and more representative of the real functionality of a complete automatic transcription system.

Overall, the experimental results validate the effectiveness of the proposed end-to-end framework for automatic swara segmentation and classification. More importantly, they show that choosing a suitable feature representation is crucial for getting good performance. In the case of monophonic flute melodies, the Constant-Q Transform (CQT) was found to be the most effective feature for both segmentation and classification tasks, for which the proposed framework yielded the best results without the need for manually labeled continuous recordings.

Chapter 6

LIMITATIONS

Although the proposed framework achieved promising results for automatic swara segmentation and classification, several limitations remain.

First, the experiments were conducted exclusively using flute recordings. While the proposed synthetic data generation strategy and end-to-end pipeline were designed for monophonic musical instruments in general, the present study only validates the framework on flute audio. Consequently, although the methodology is expected to be applicable to other monophonic instruments, such as the violin, recorder, clarinet, or saxophone, additional experiments are required to verify its generalization capability.

Secondly, the proposed system has been designed to take care of only seven fundamental swaras (Sa, Re, Ga, Ma, Pa, Dha, Ni). In this work, musical ornamentations, such as Meend, Gamak, Murki, and Kan Swara, found in Indian classical music were omitted from the current dataset and evaluation, and previous studies indicate that the ornamentation and the melody context have a significant effect on the acoustic realization of a note [39, 40]. The framework has therefore not yet been evaluated on more complex performance techniques.

Thirdly, even though the synthetic melody generation process added variations by using several kinds of transition style and digital signal processing augmentations, synthetically generated melodies cannot completely mimic all the traits which exist in actual performances. Acoustic characteristics in the database may not be fully represented due to variations in the style of the performer, breathing technique, articulation, and recording environments.

Last, the proposed scheme is based on monophonic audio; that is, it is based on the assumption that there is at most one note playing at any given time. The approach presented here cannot then be directly applied to polyphonic music or recordings of polyphonic music with multiple simultaneous instruments, without making substantial changes to both the segmentation and classification components.

Chapter 7

FUTURE WORK

The proposed framework provides a foundation for automatic swara segmentation and classification using synthetically generated melodies. Although the obtained results are promising, several directions exist for extending and improving the proposed system.

The most immediate future work is to evaluate the proposed pipeline on other monophonic musical instruments. The study was conducted mainly on recordings of flute, but the methodology is independent of instrument-specific features. By replacing the isolated note recordings with those of other monophonic instruments, such as the violin, recorder, clarinet, saxophone, or similar instruments, the same synthetic melody generation pipeline can be employed to create automatically annotated datasets. Further validation experiments using other instruments could be more convincing for the generalizability of the proposed framework.

Another important extension is the use of more expressive musical techniques that are used in Indian classical music. The present work only takes into account the seven basic swaras; however, ornamentation, such as Meend, Gamak, Murki and Kan Swara, is often used in practical performance. Incorporating these expressive elements into this synthetic generation of the melody would allow the proposed framework to work on more realistic and musically complex performance.

Although the proposed synthetic dataset successfully reduced the need for manually annotated continuous recordings, future work may further enhance the realism of the generated melodies. More complex transition modeling, individual player playing style, breathing effects, expressive dynamics and other acoustic augmentation might be adapted to ensure a closer approximation to real-life performances. Such improvements would further narrow the gap between synthetic and real-world musical recordings.

Since the synthetic melody generation process is fully automated, future research could investigate generating substantially larger datasets with thousands or even millions of automatically annotated melodies. These large-scale synthetic data sets can help train more sophisticated deep learning models without the need for intensive manual annotation.

Advanced deep learning architectures can also be explored in future research. While this work employed CNN with Attention and CRNN with Attention models, recent sequence modeling techniques such as Transformer-based architectures [6], Conformers [53], and self-supervised audio representation models [54] could potentially improve both segmentation and classification performance, particularly for longer and more complex musical recordings.

Finally, the proposed system can be evolved towards an eventual automatic music transcription system operating in real time. The ability to integrate segmentation, swara classification, notation generation and visualization into the same application would allow for an automatic

analysis of monophonic performances by musicians, educators and researchers. This system could facilitate music education, digital archiving of traditional performances and future music information retrieval applications.

Chapter 8

CONCLUSION

Automatic swara segmentation and classification play an important role in music information retrieval, digital music analysis, and computer-assisted music education. The task of building effective automatic transcription systems, however, is hindered by the lack of good quality accurately annotated continuous music recordings. Manual annotation of note boundaries is a labor-intensive and time-consuming process, making it difficult to construct large datasets for training deep learning models. Motivated by this challenge, this research proposed an end-to-end framework that minimizes the dependence on manually annotated continuous recordings by utilizing synthetically generated melodies with automatically created ground-truth annotations.

The proposed framework consists of four major stages: dataset preparation, feature extraction, swara segmentation, and swara classification. An individual swara dataset containing recordings of the seven fundamental swaras was first collected from both publicly available recordings and self-recorded flute performances. These isolated recordings were then processed to create a synthetic set of melodies by randomly selecting notes, modelling the transitions between them and making multiple digital signal processing enhancements. The process of generating melodies automatically recorded temporal boundaries of each note that was inserted, so there was no need to make manual annotations on a frame-by-frame basis.

To evaluate the proposed framework, six different acoustic feature representations were investigated, namely Log-Mel, Pitch, RMS, Log-Mel + Pitch, Log-Mel + Pitch + RMS, and Constant-Q Transform (CQT). In all experiments, identical segmentation and classification architectures have been used and differences in performance are due to the particular feature representation. Experimental results demonstrated that the Constant-Q Transform provided the most effective representation for monophonic flute melodies, enabling the proposed framework to achieve the highest overall performance, with an end-to-end segmentation accuracy of 92.86% and a note-level classification accuracy of 97.80% on correctly segmented melodies.

Apart from the results of the work performed, the main advantage of the work in this thesis is that it serves to show that swara segmentation and classification can be achieved without the need for hundreds of hours of manually labelled continuous recordings. This proposed approach for synthetic melody generation not only significantly decreases the amount of annotations but also ensures accurate ground-truth data for training and evaluating the model. This allows for a practical and scalable solution, when only a few or no annotated datasets exist for developing automatic music transcription systems.

The general approach developed in this work is not specific to the flute; however, the work was tested only using flute recordings. Even though this pipeline relies on separate recordings of notes, it may be possible to transfer this approach to other monophonic musical instruments

by using different source recordings, while keeping the rest of the pipeline intact. This flexibility provides a promising foundation for future research in automatic music transcription and music information retrieval.

In conclusion, this thesis demonstrates that combining synthetic dataset generation with deep learning-based segmentation and classification models offers an effective solution for automatic swara identification. The proposed framework not only achieves promising performance but also presents a practical approach for reducing manual annotation requirements, making it a valuable contribution toward scalable and data-efficient research in computational music analysis.

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